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(54) **MEMORY DEVICE AND METHOD FOR FORMING THE SAME**

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(52) **U.S. Cl.**

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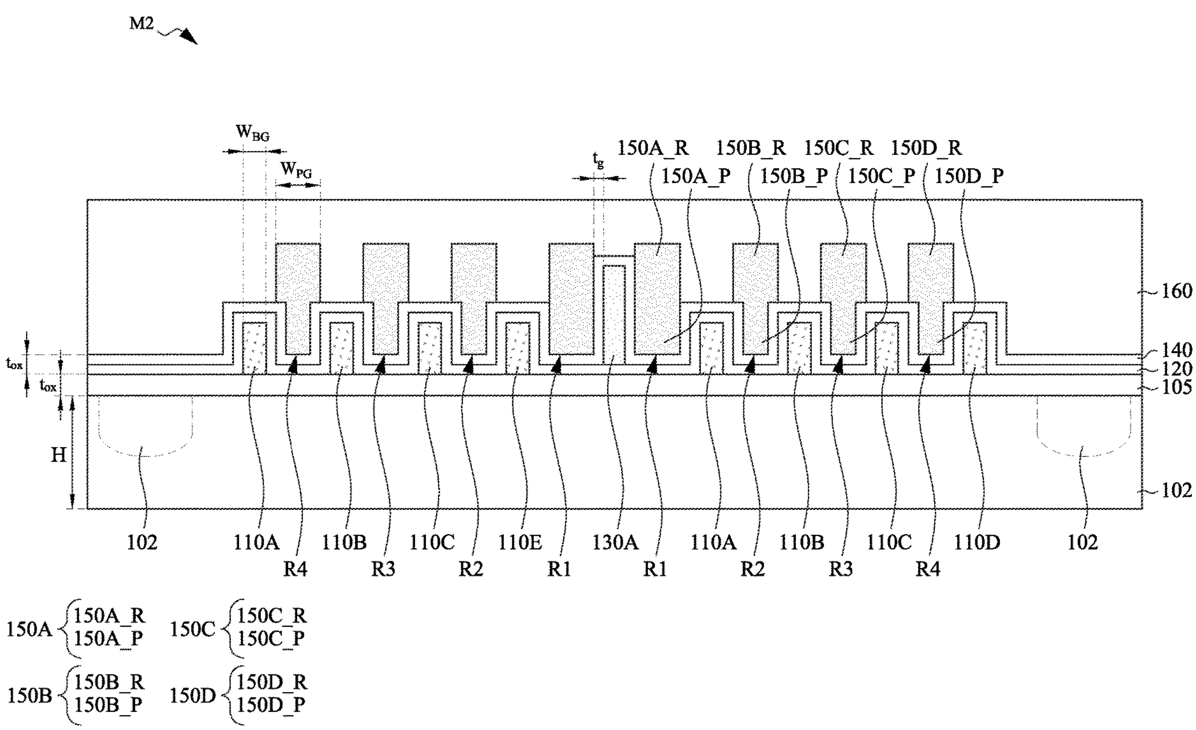
(57) **ABSTRACT**

A memory device includes a substrate. A gate dielectric layer is over the substrate. A first plunger gate structure and a second plunger gate structure are over the gate dielectric layer, wherein in a top view the first and second plunger gate structures are arranged concentrically, and wherein each of the first and second plunger gate structures comprises a ring structure and plunger gates extend downwardly from the ring structure. Barrier gates are over the gate dielectric layer, wherein in a cross-sectional view each of the plunger gates of the first and second plunger gate structures is laterally between adjacent two of the barrier gates.



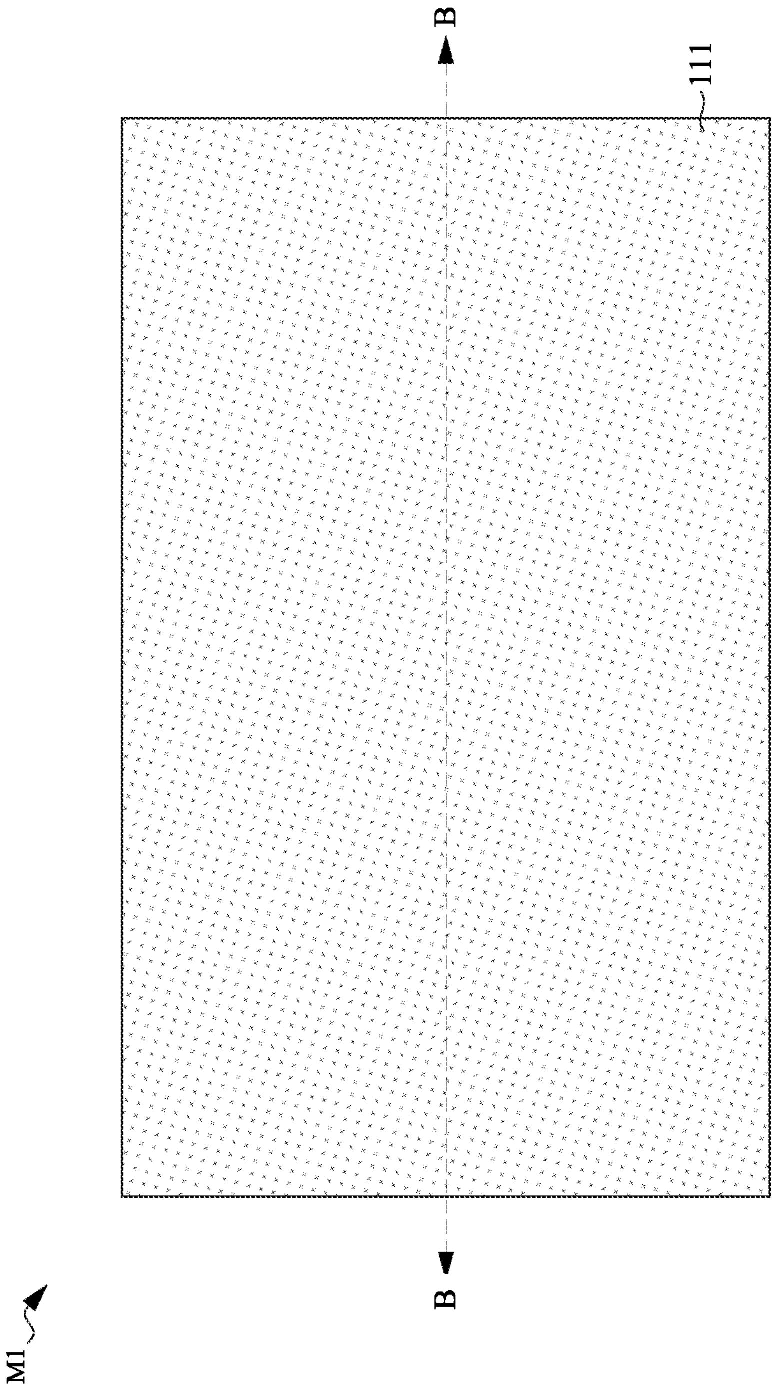


Fig. 1A

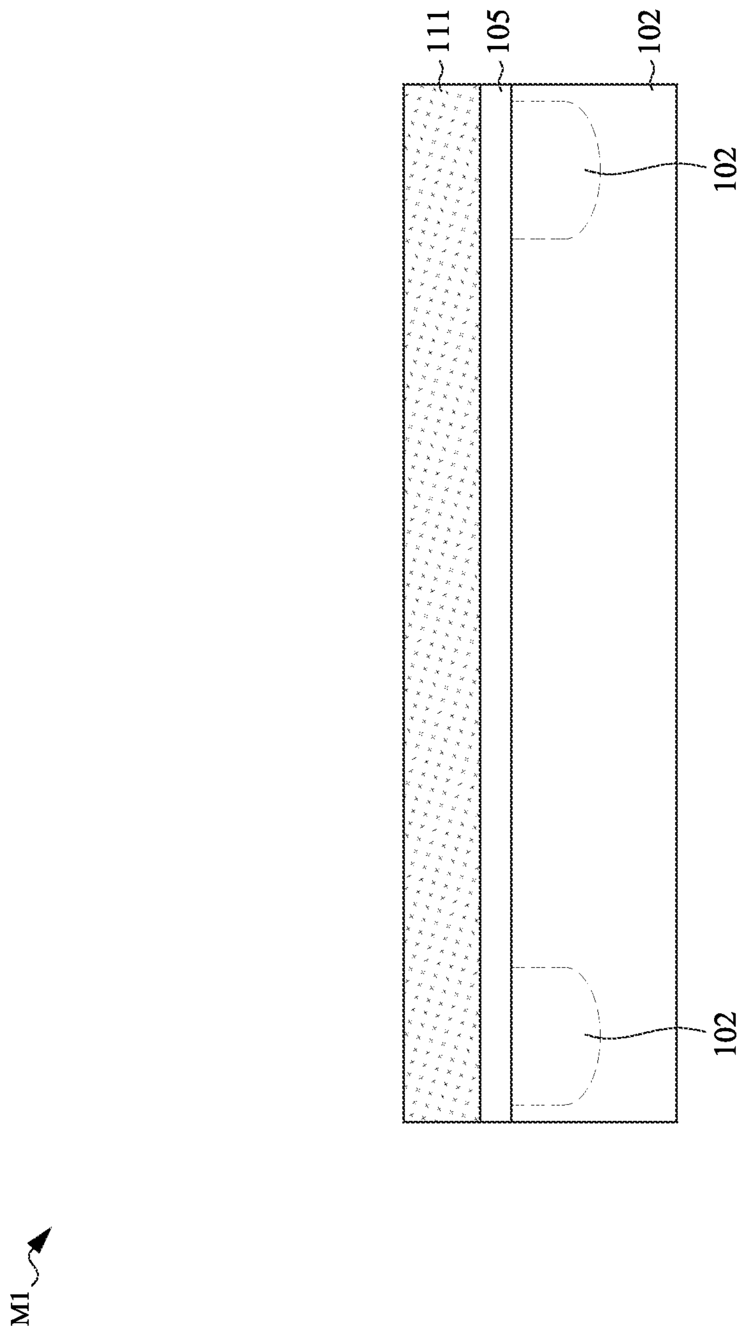


Fig. 1B

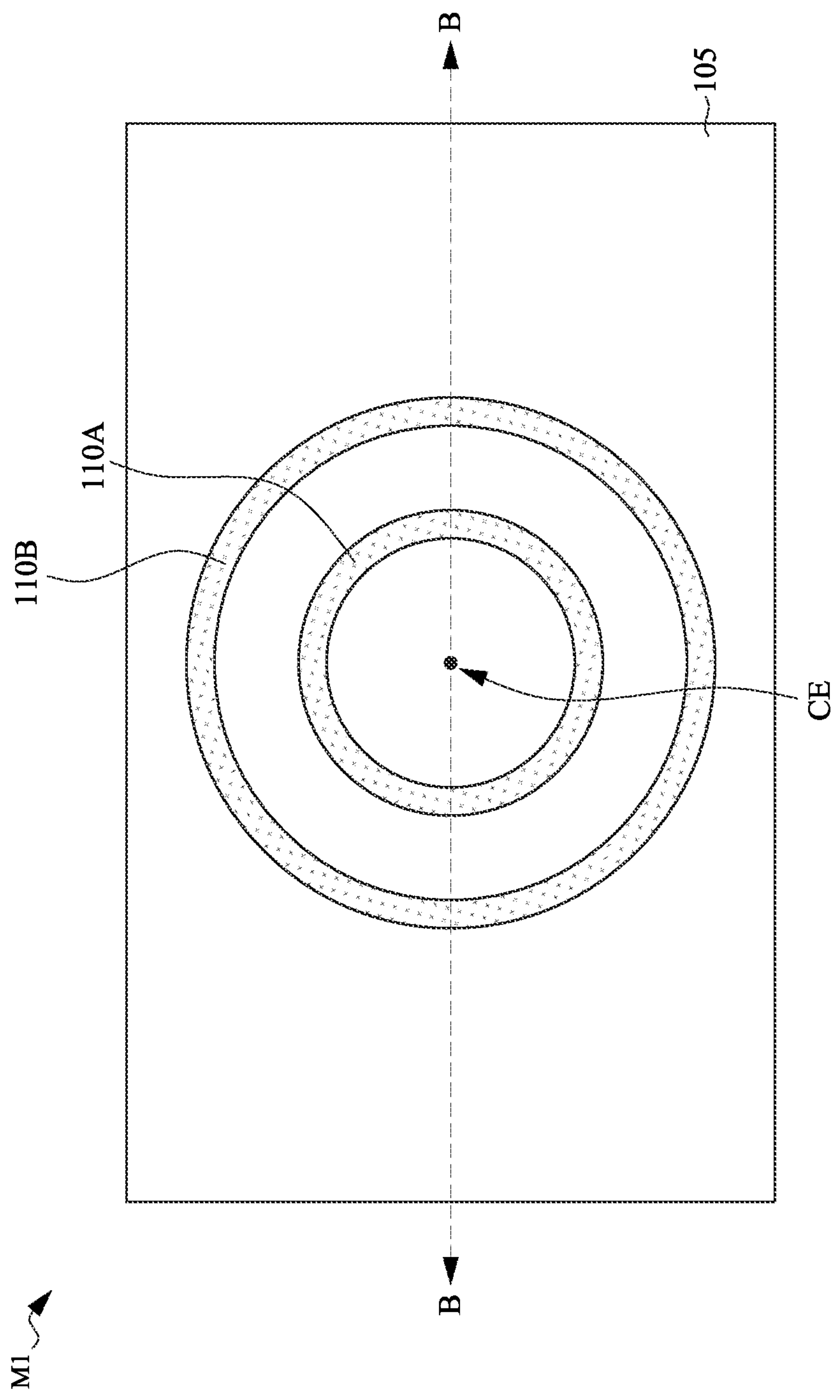


Fig. 2A

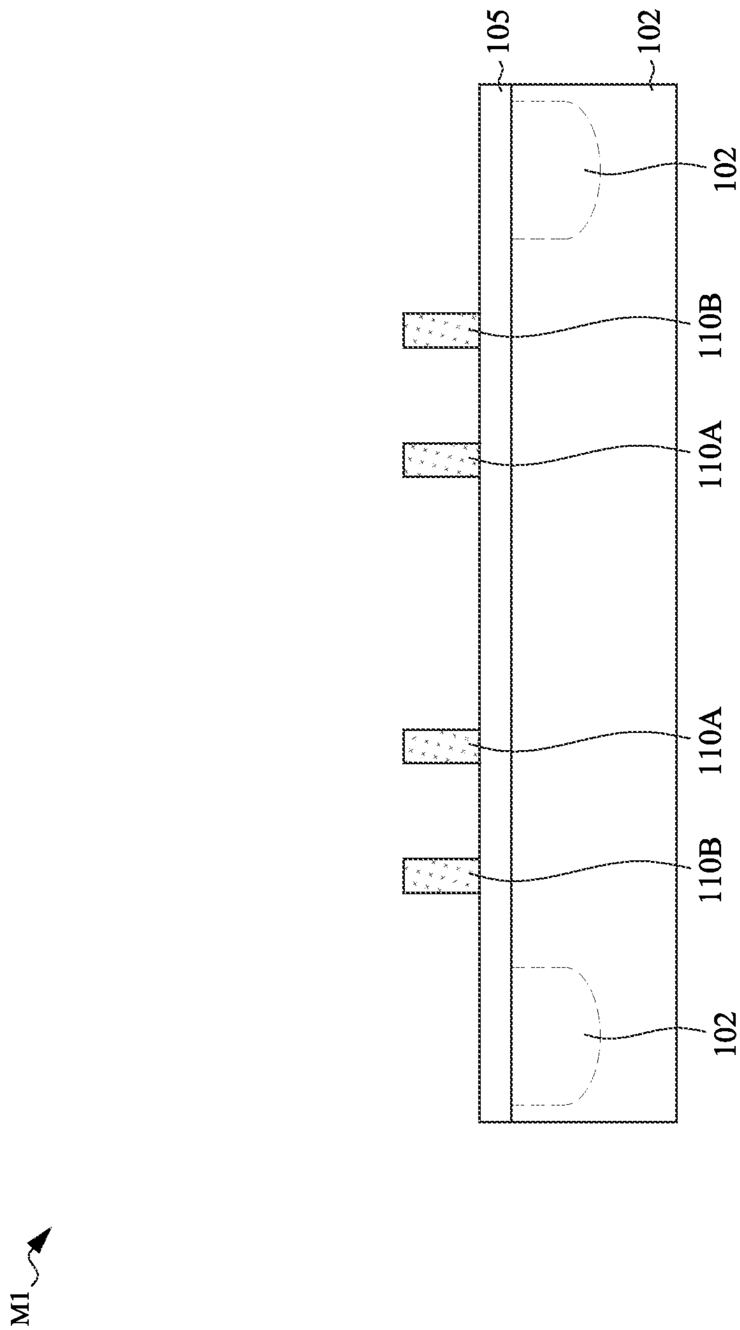


Fig. 2B

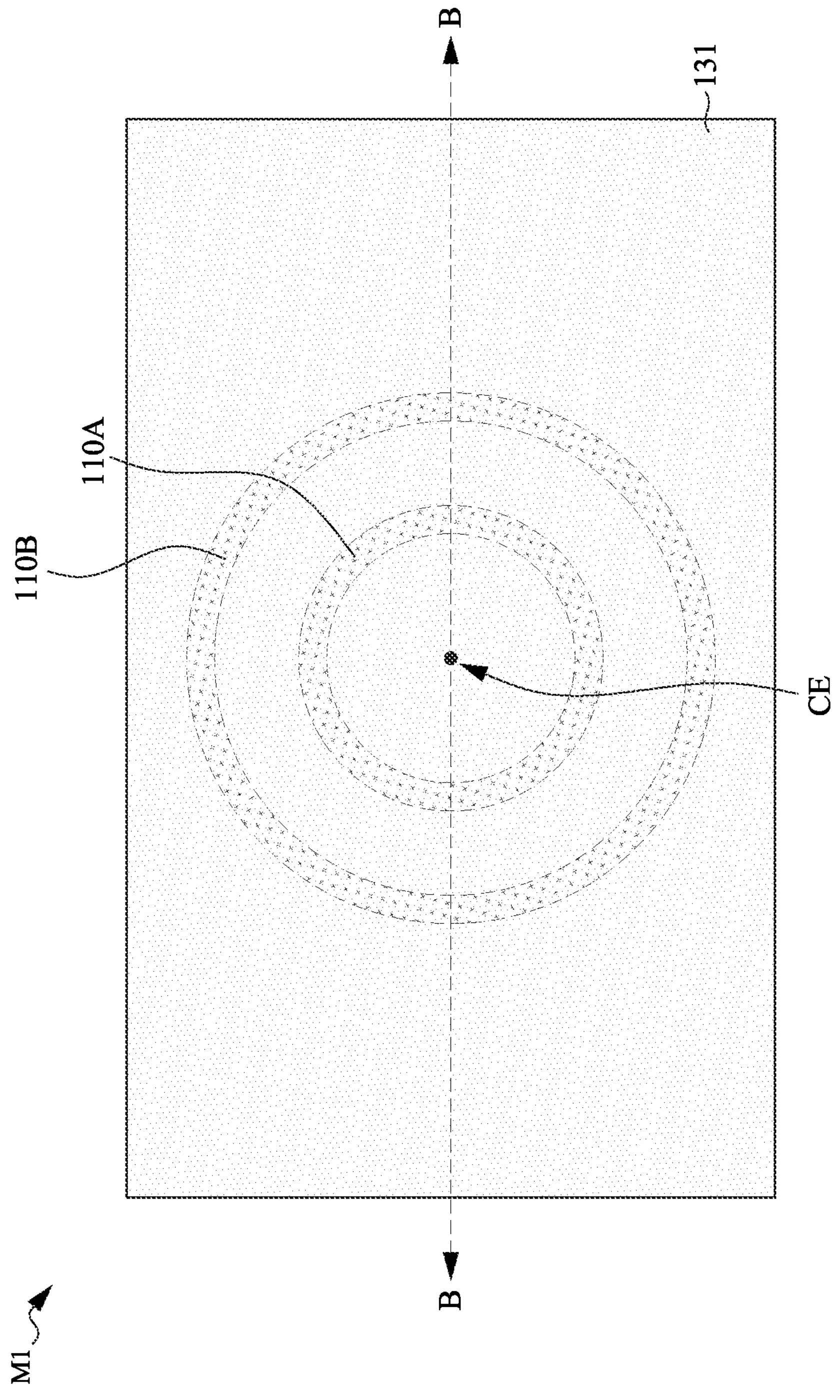


Fig. 3A

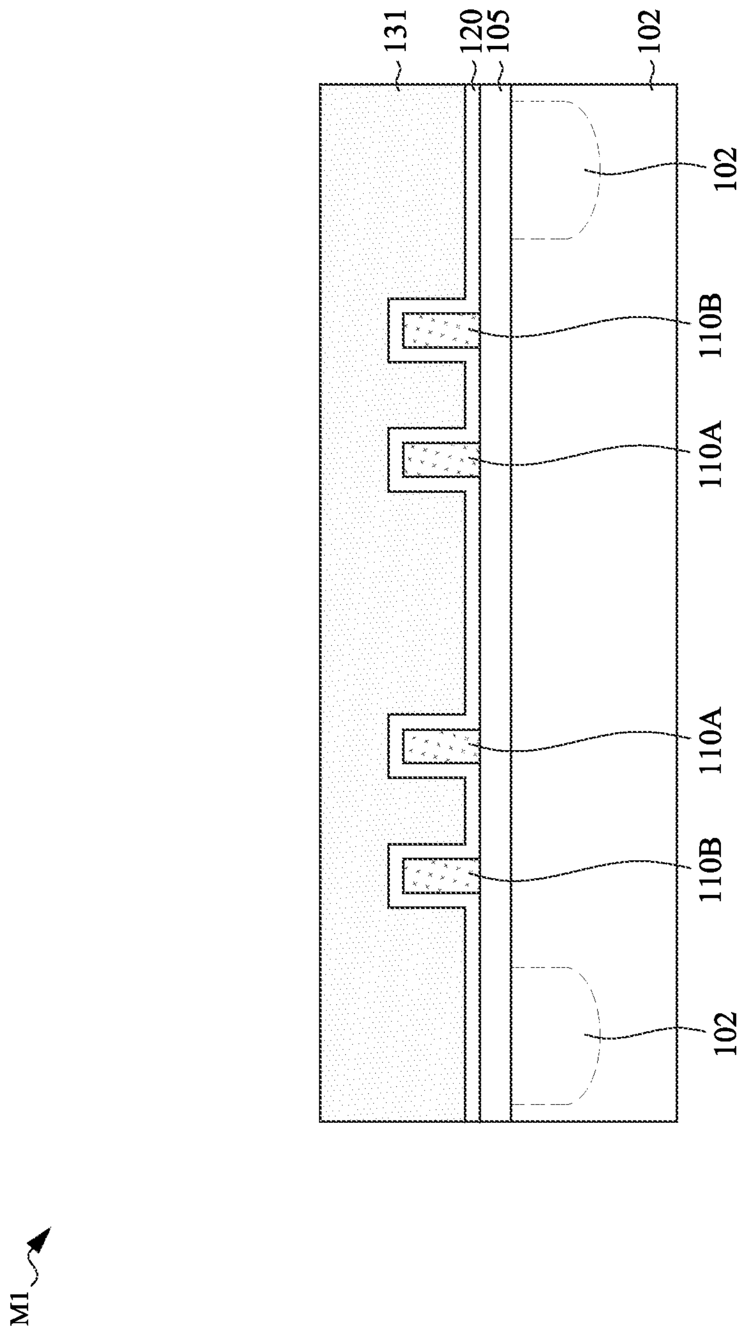


Fig. 3B

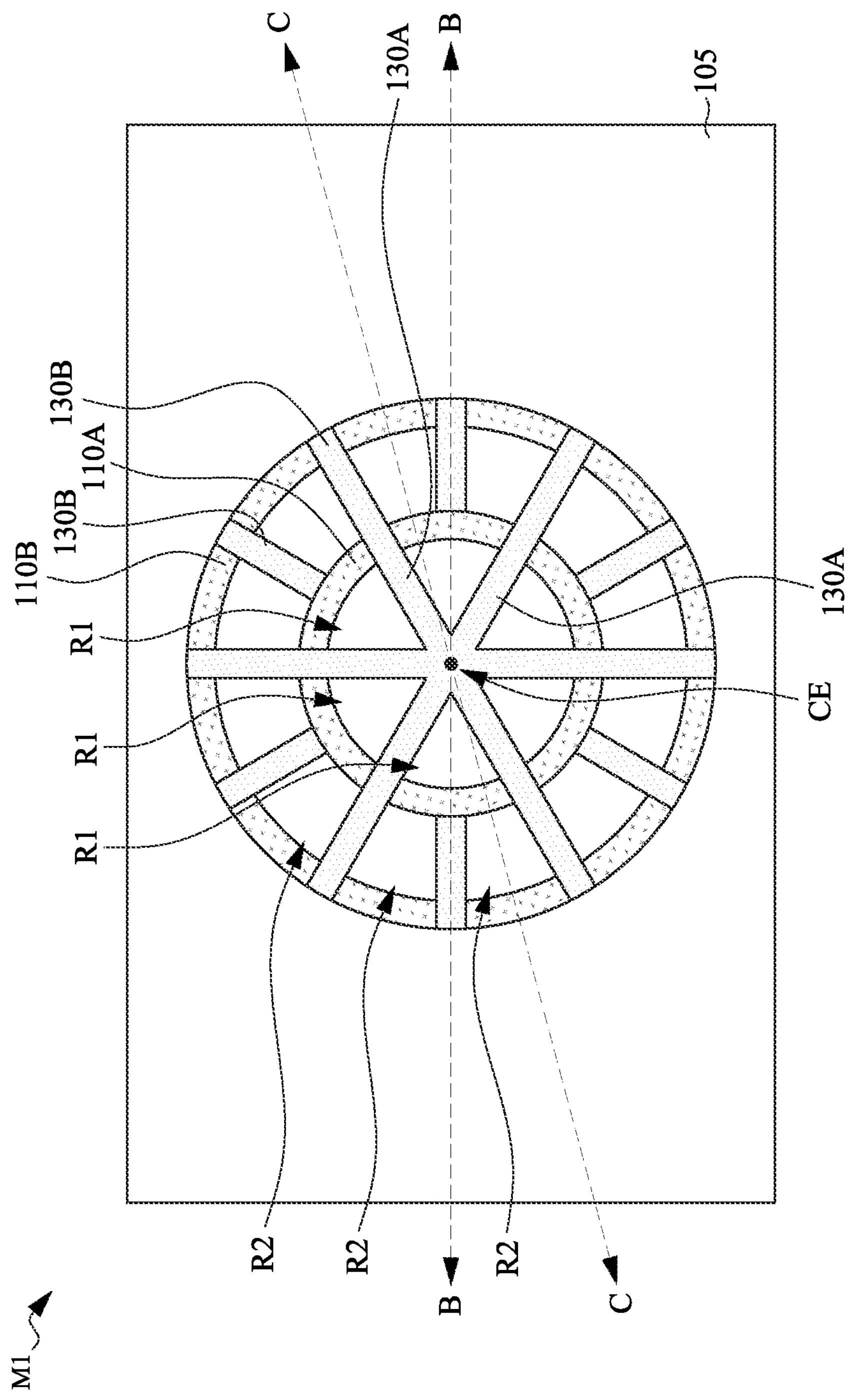


Fig. 4A

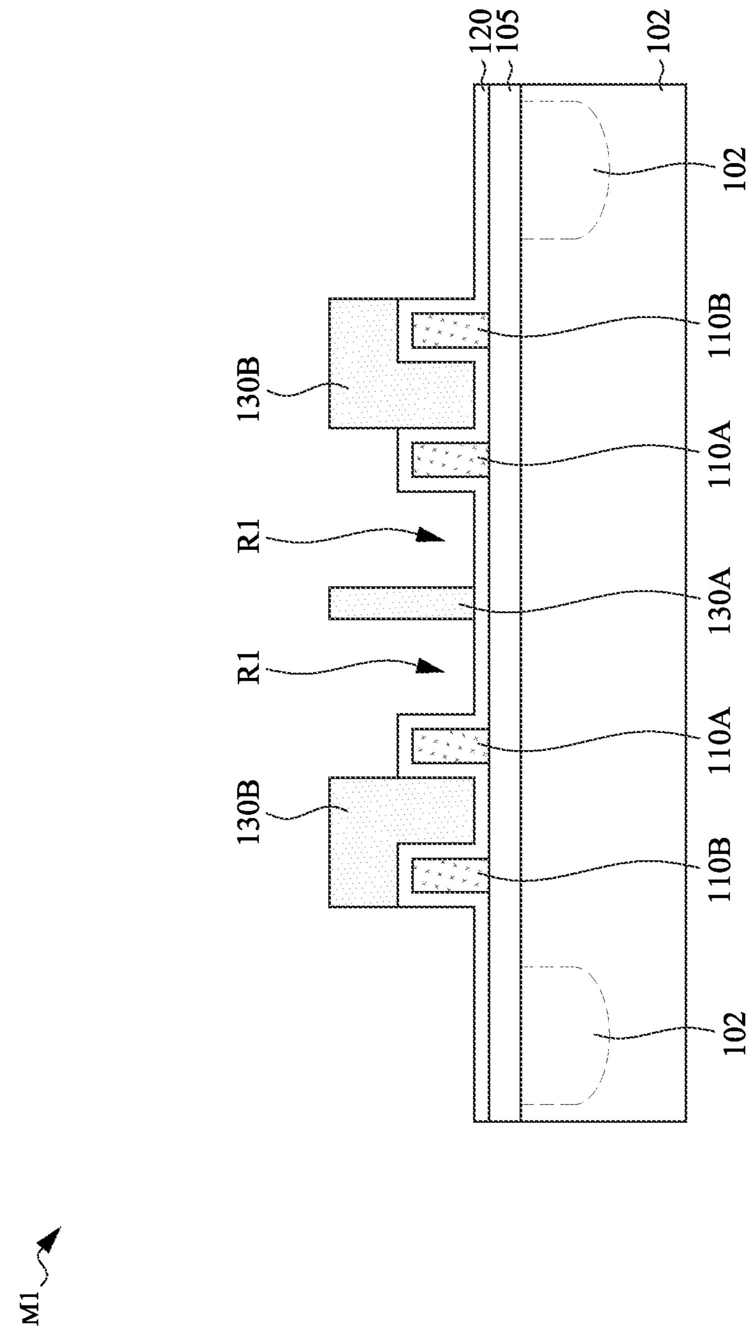


Fig. 4B

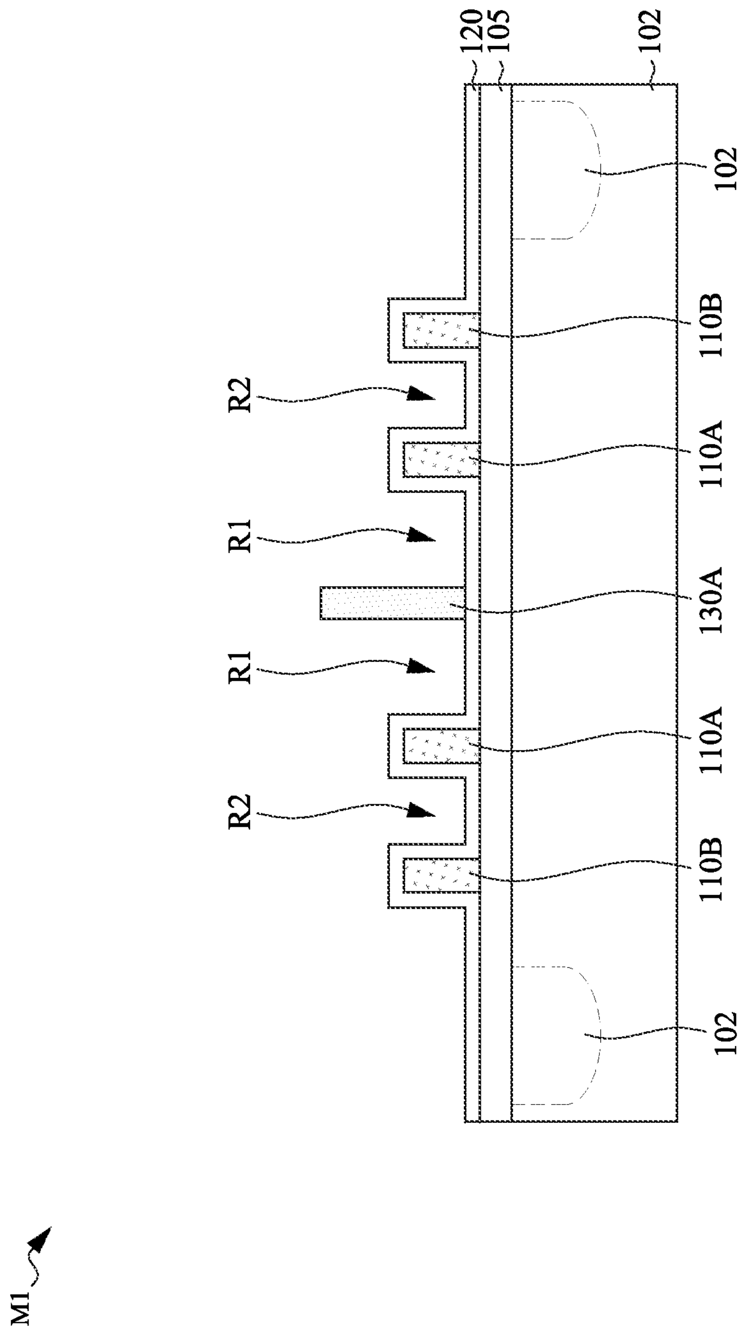


Fig. 4C

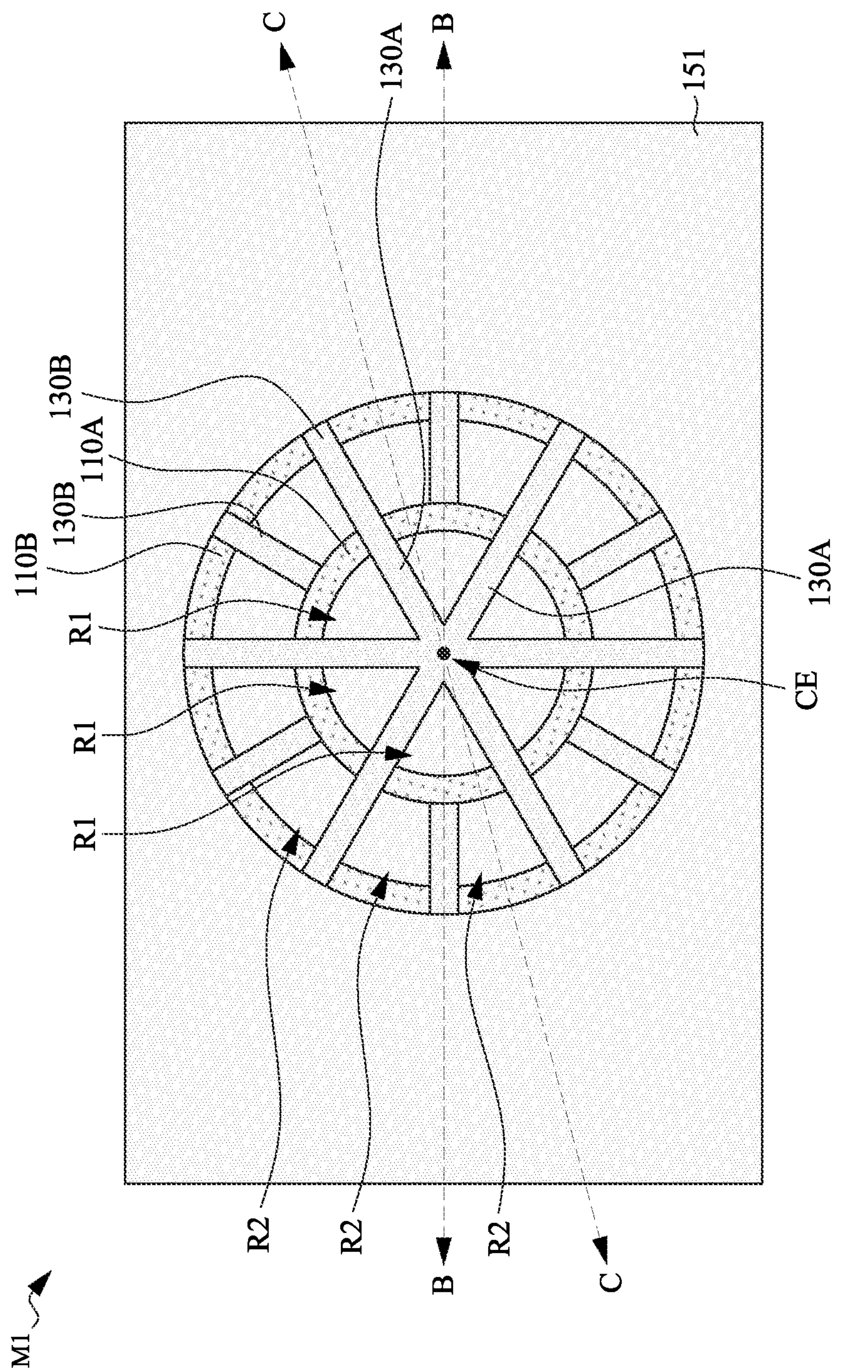


Fig. 5A

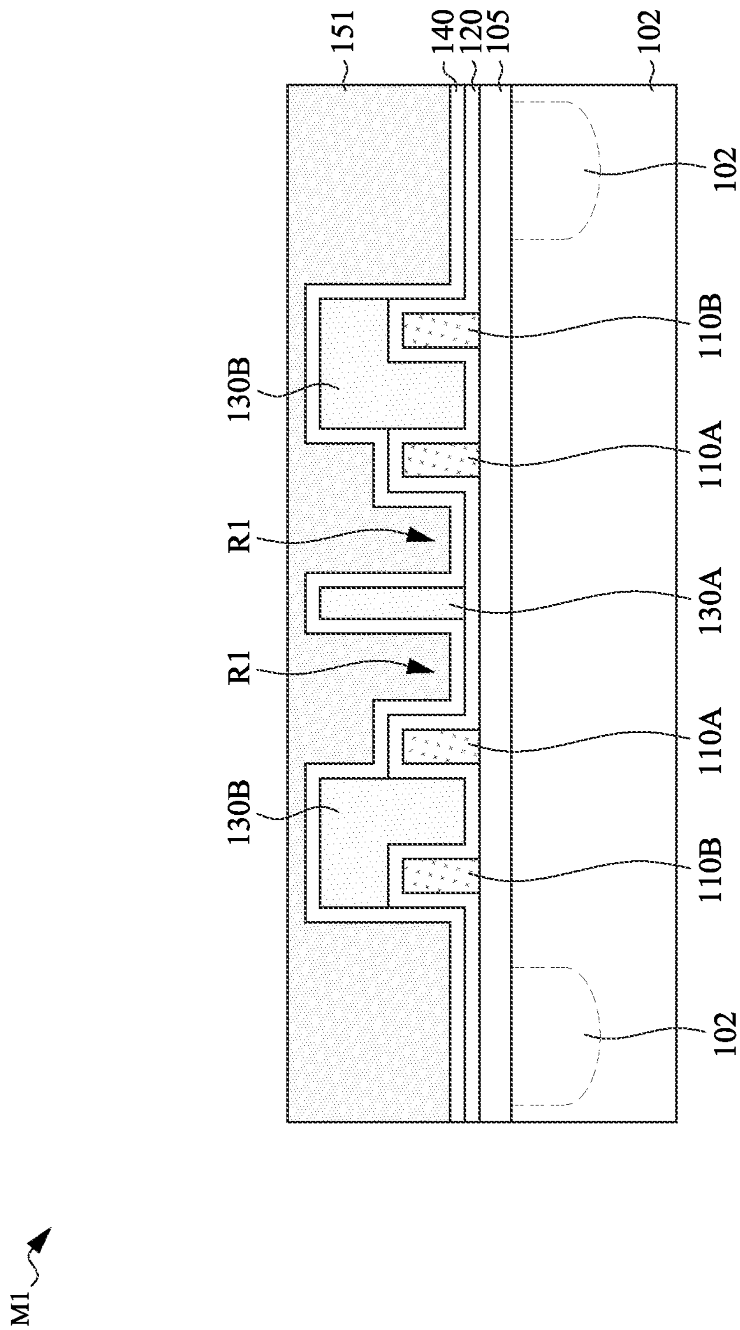

151
140
120
105
102
102
110B
110A
130A
110A
110B
102
130B
R1
R1
130B
M1

Fig. 5B

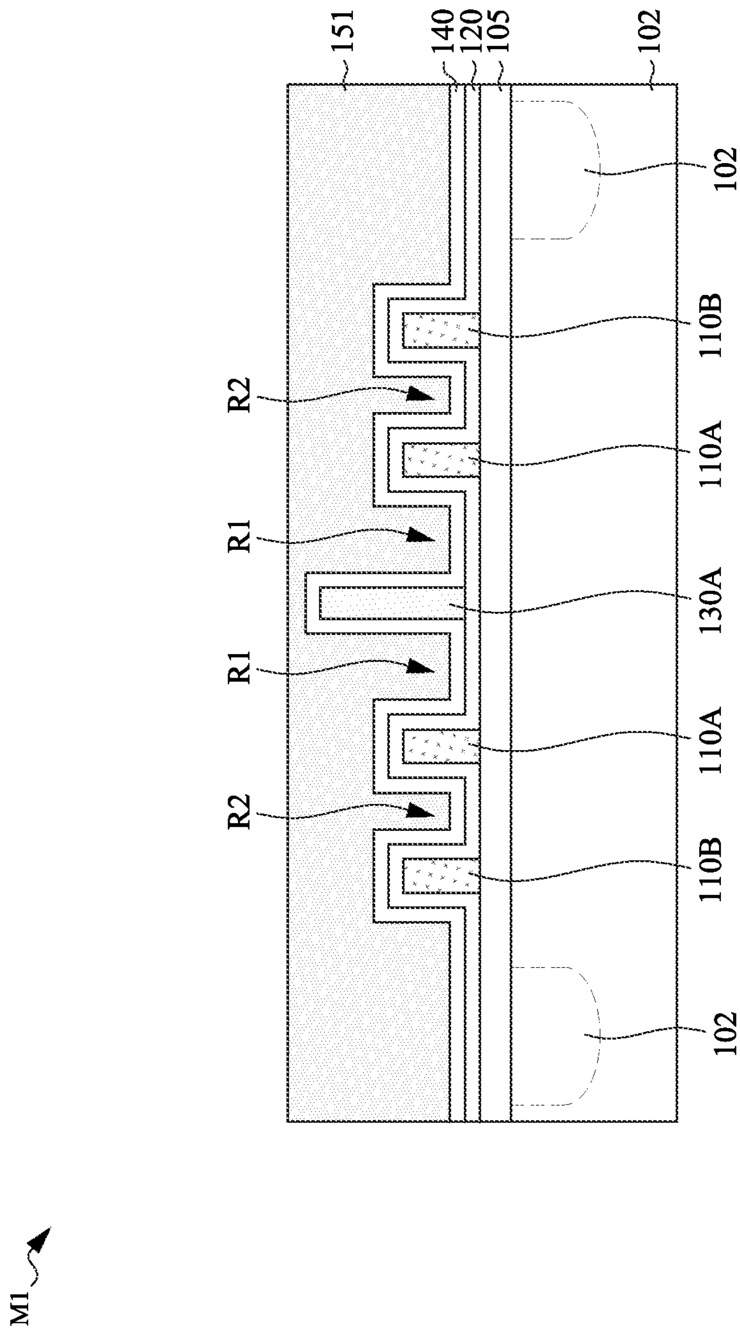

151
140
120
105
102
102
110B
110A
130A
110A
110B
102
R2
R1
R1
R2
M1

Fig. 5C

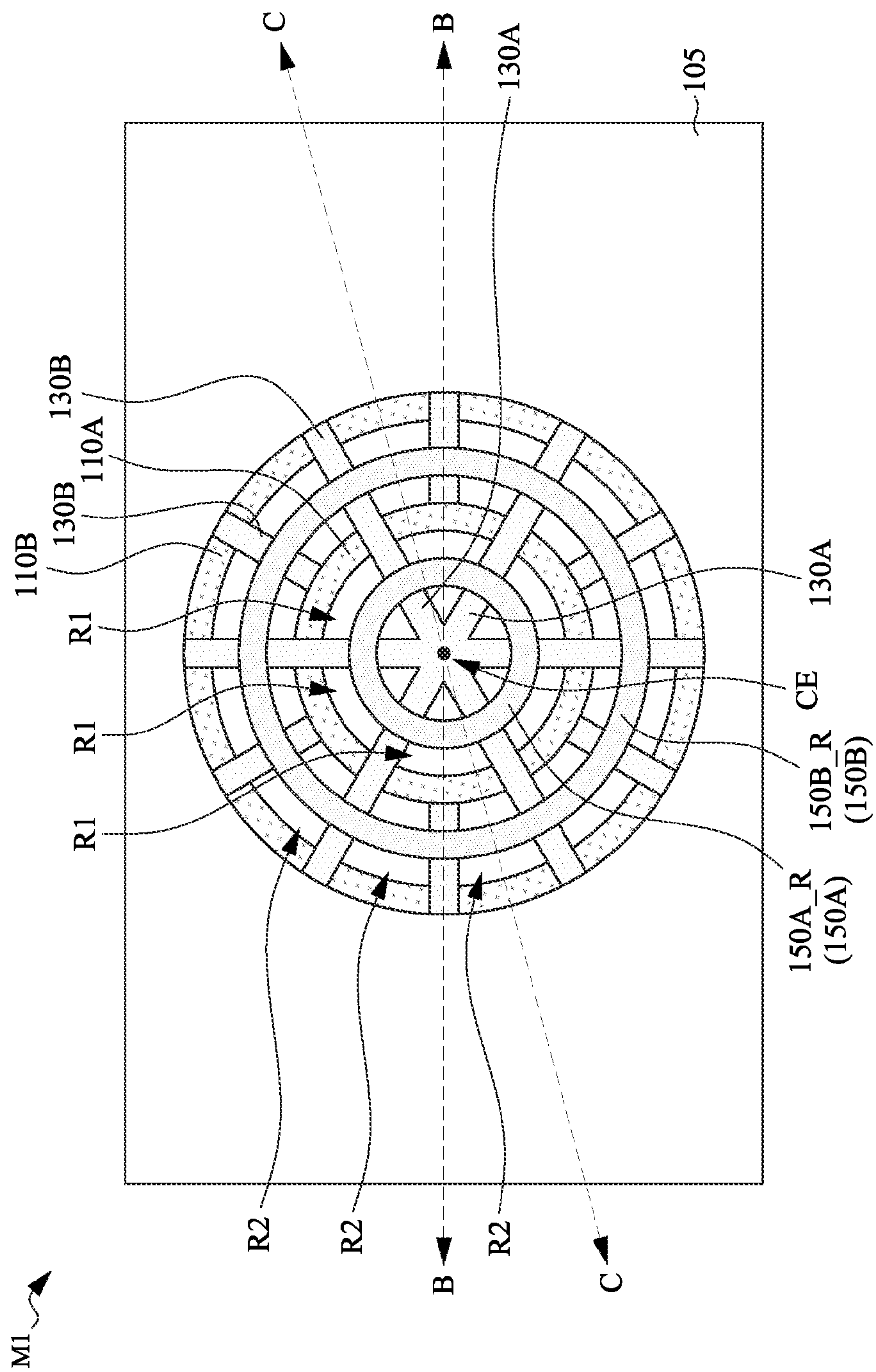

C
B
130A
105
130B
110A
130B
110B
R1
R1
R1
130A
CE
150B_R
(150B)
150A_R
(150A)
R2
R2
R2
B
C
M1

Fig. 6A

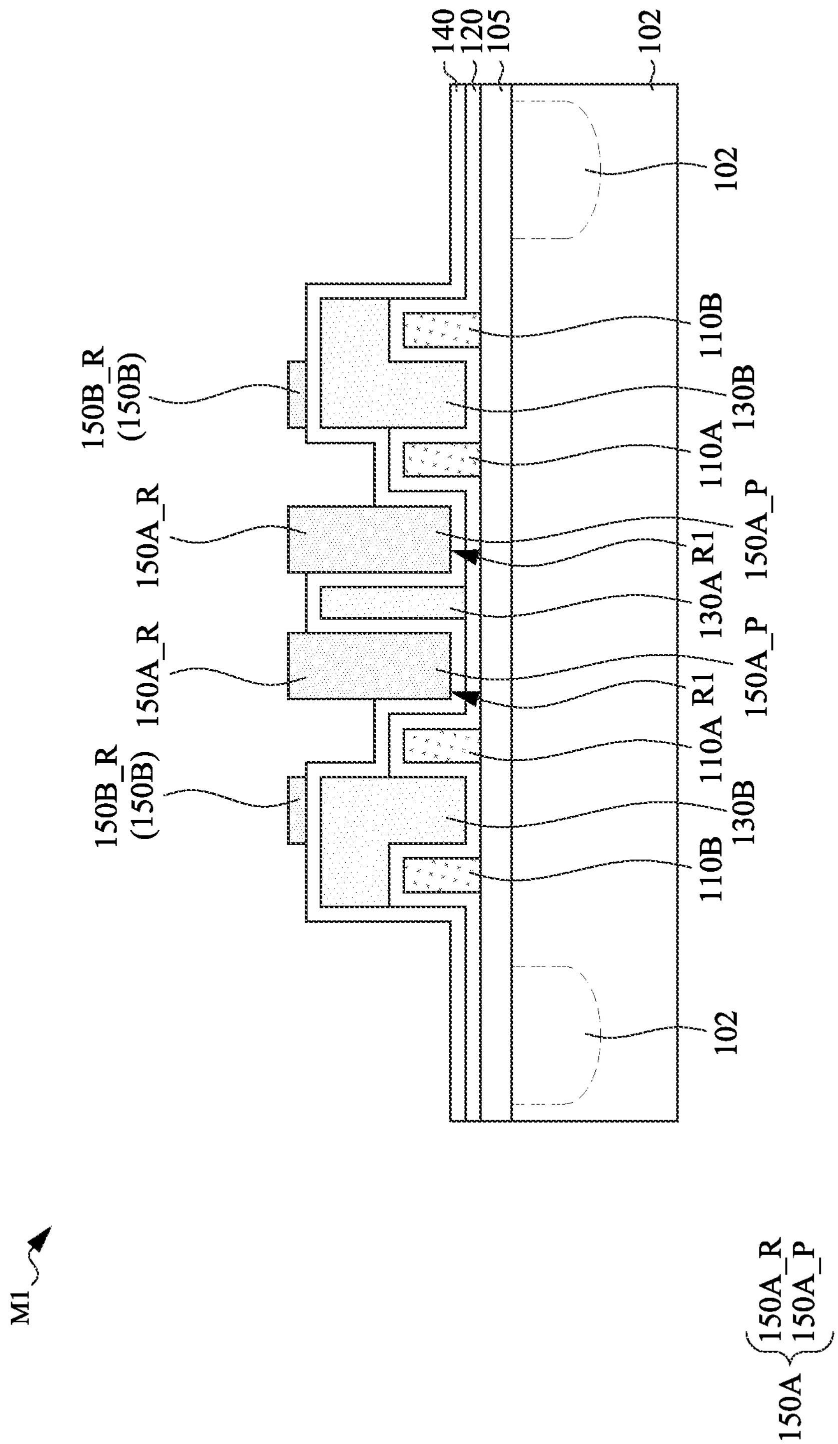

140
120
105
102
102
110B
130B
150B_R
(150B)
110A
150A_R
150A_P
R1
130A
150A_R
150A_P
R1
110A
150B_R
(150B)
130B
110B
102
M1
150A
150A_R
150A_P

Fig. 6B

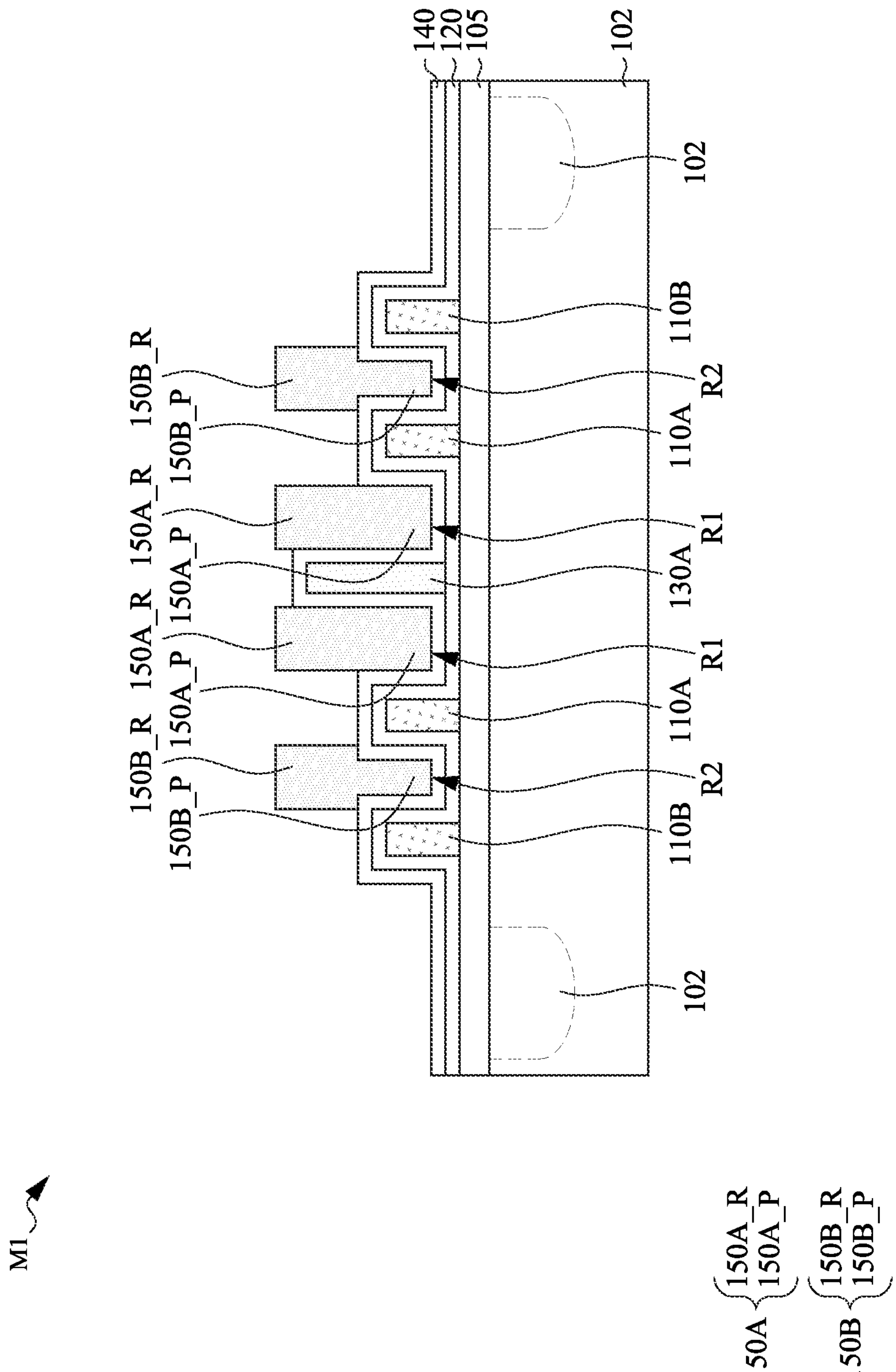

140
120
105
102
102
110B
150B_R
150B_P
R2
110A
150A_R
150A_P
R1
130A
150A_R
150A_P
R1
110A
150B_R
150B_P
R2
110B
102
M1
150A
150A_R
150A_P
150B
150B_R
150B_P

Fig. 6C

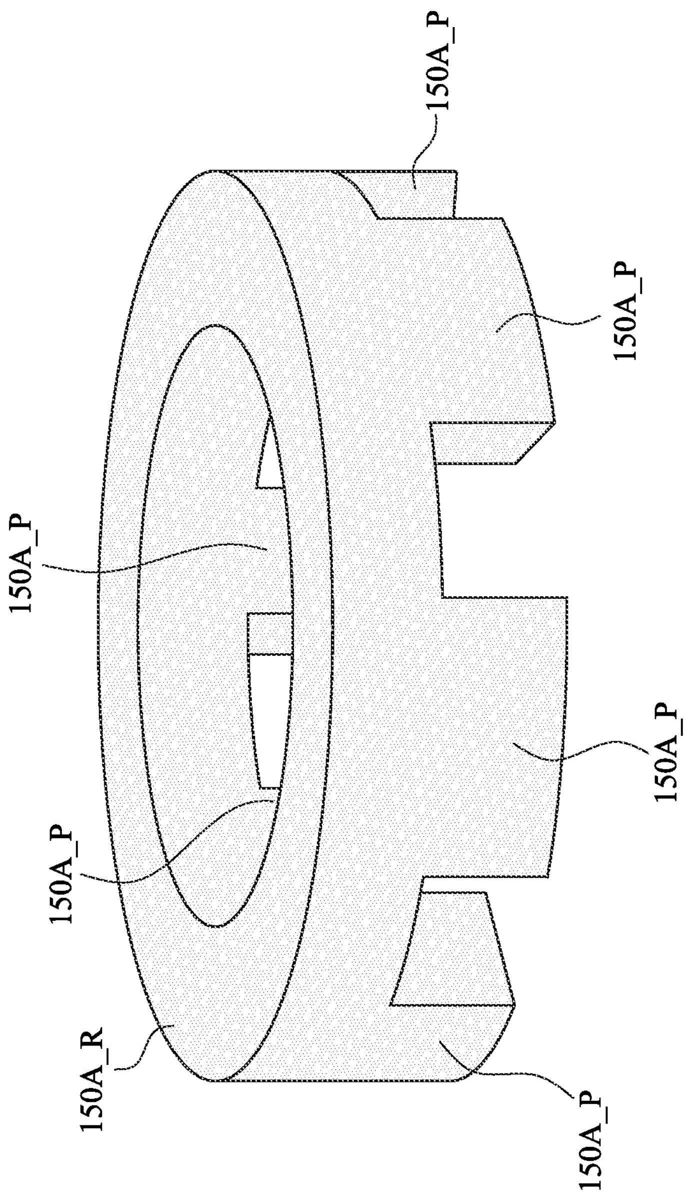


Fig. 6D

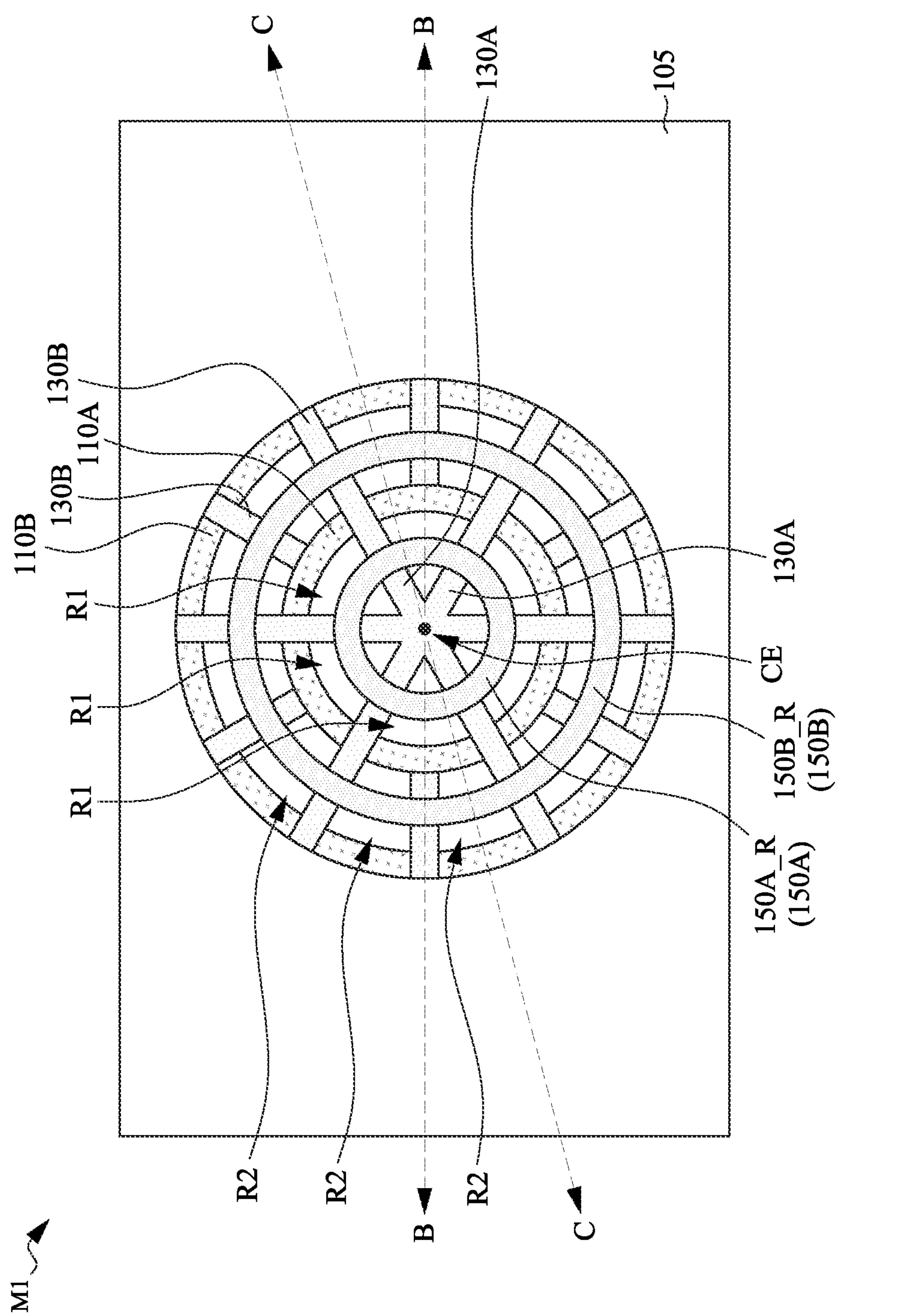

C
B
130A
105
130B
110A
130B
110B
R1
R1
R1
130A
CE
150B_R
(150B)
150A_R
(150A)
R2
R2
R2
B
C
M1

Fig. 7A

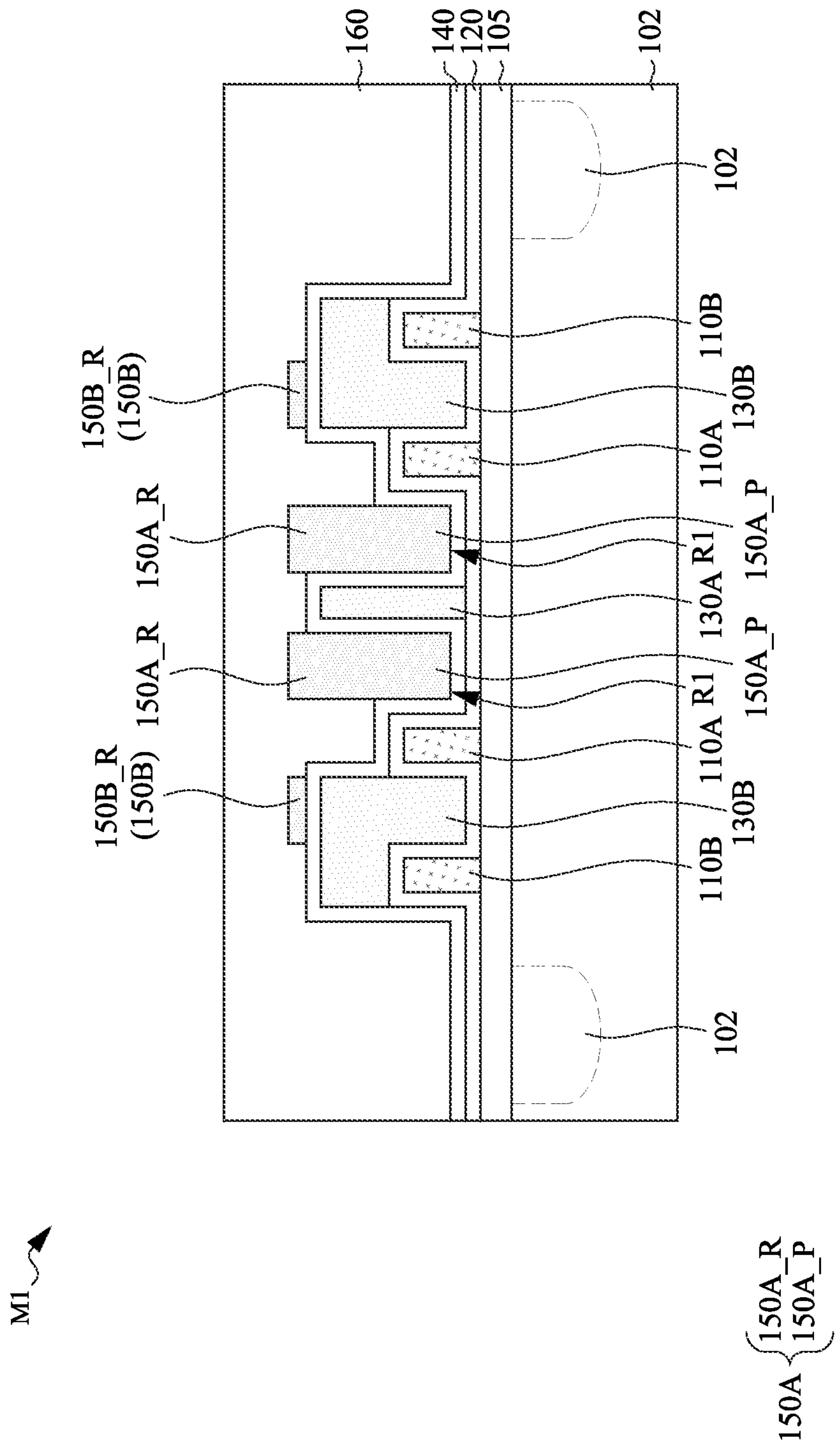
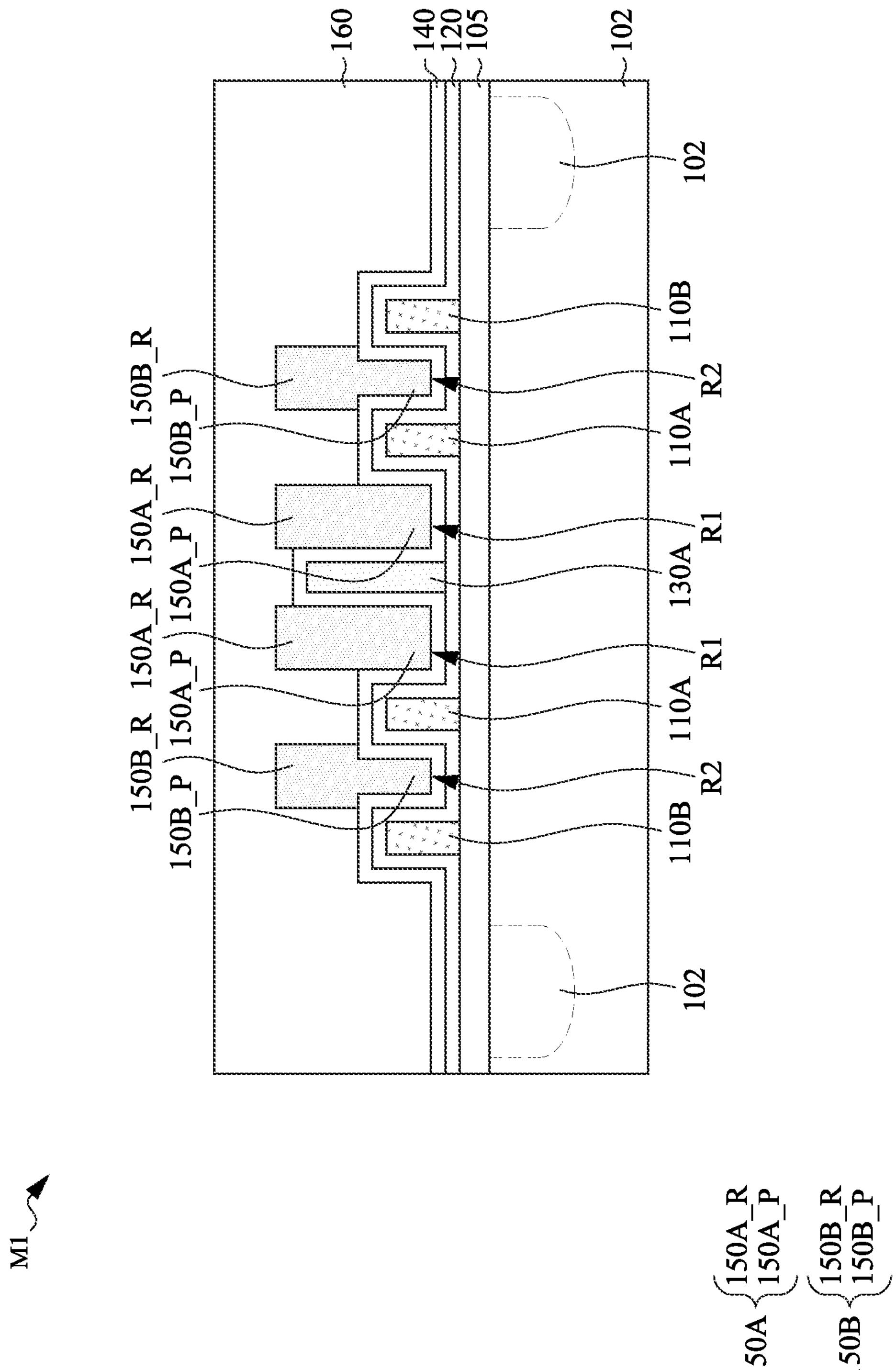

160
140
120
105
102
102
110B
130B
110A
150A_P
R1
130A
150A_P
R1
110A
130B
110B
102
150B_R
(150B)
150A_R
150A_R
150B_R
(150B)
M1
150A
150A_R
150A_P

Fig. 7B



160
140
120
105
102
102
102
110B
150B_R
R2
150B_P
110A
150A_R
R1
150A_P
130A
150A_R
R1
150A_P
110A
150B_R
R2
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110B
M1
150A
150A_R
150A_P
150B
150B_R
150B_P

Fig. 7C

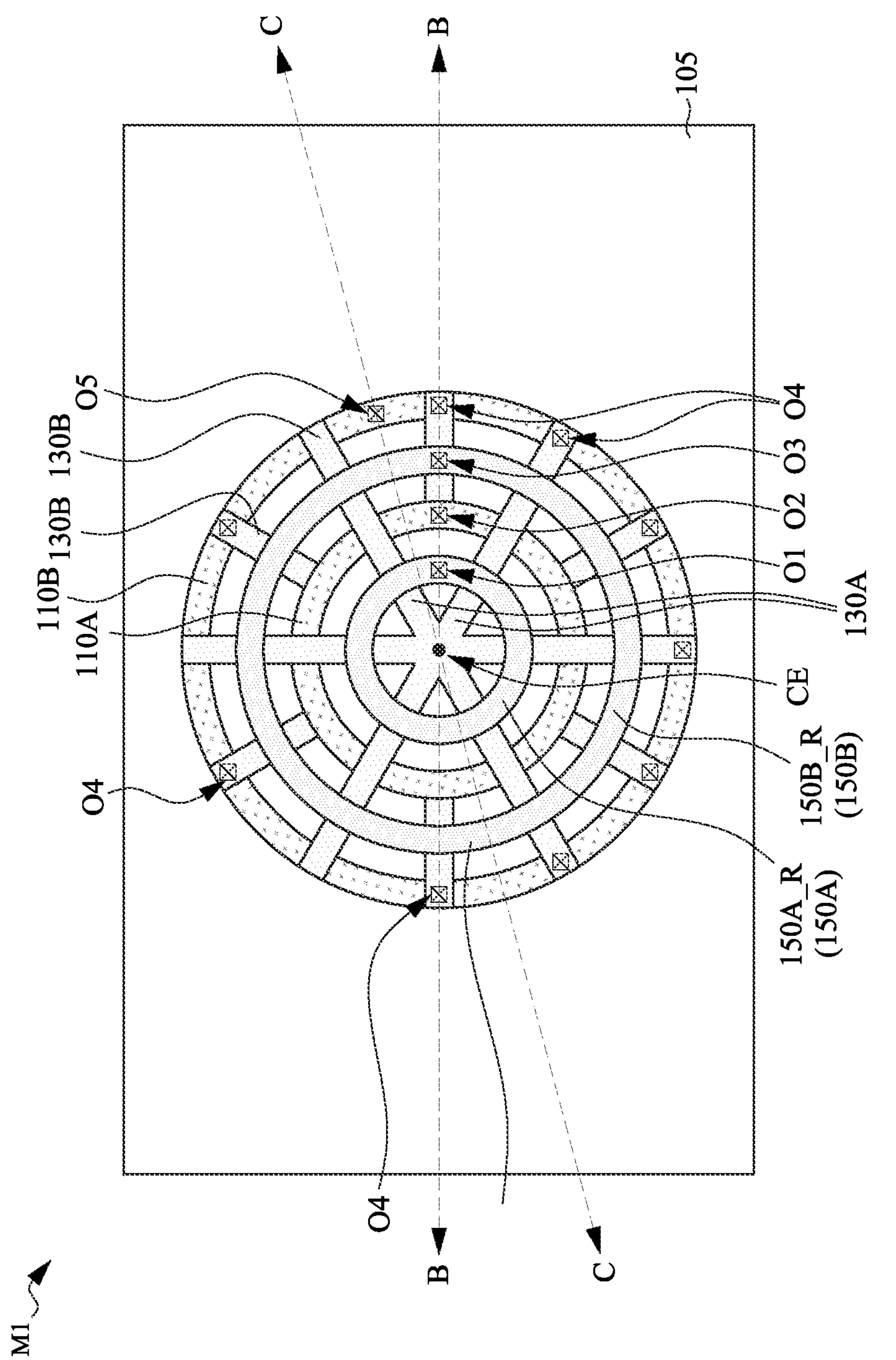

M1
105
B
B
C
C
O5
130B
130B
110B
110A
O4
O3
O2
O1
130A
CE
150B_R
(150B)
150A_R
(150A)
O4
O4

Fig. 8A

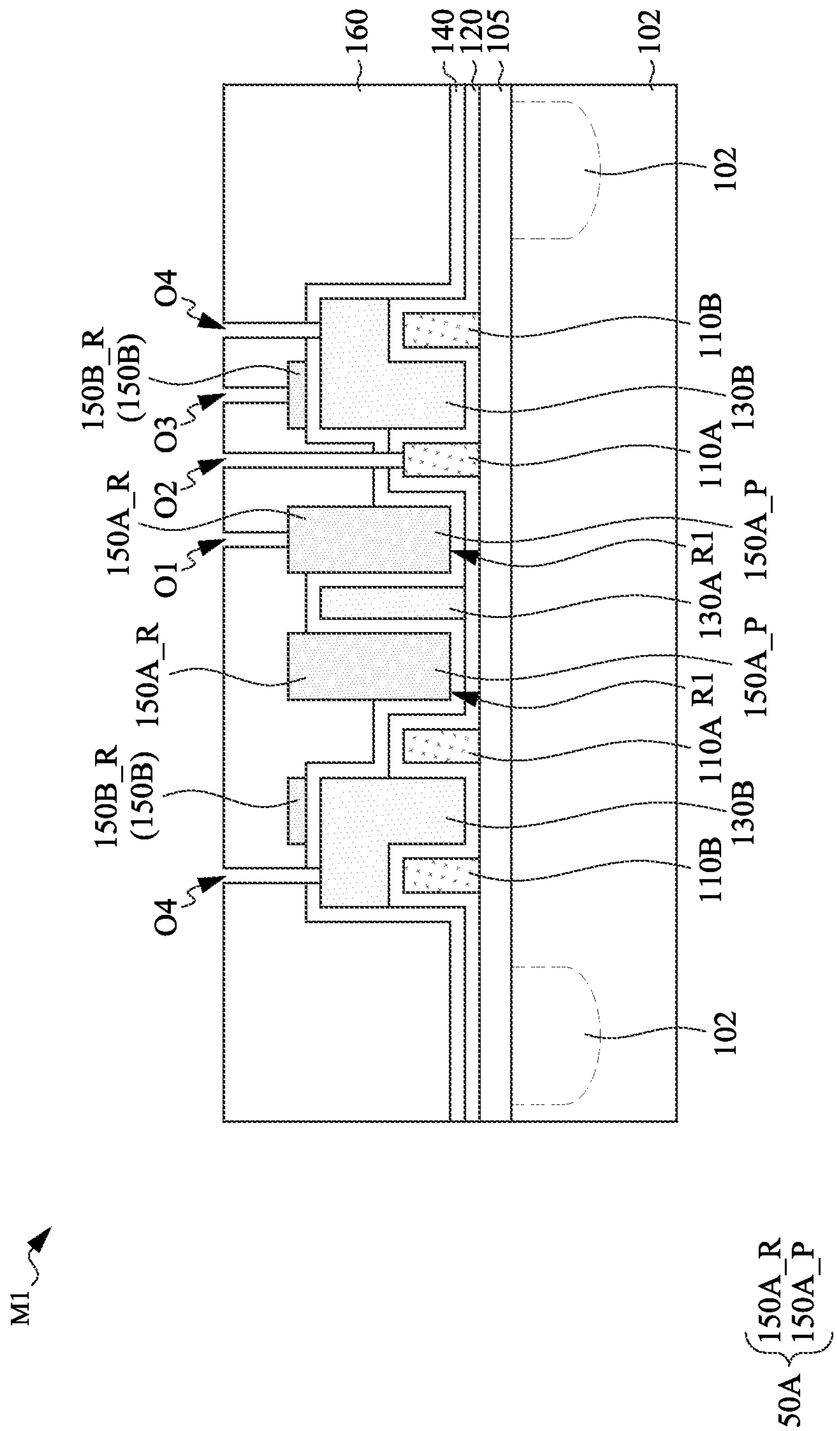
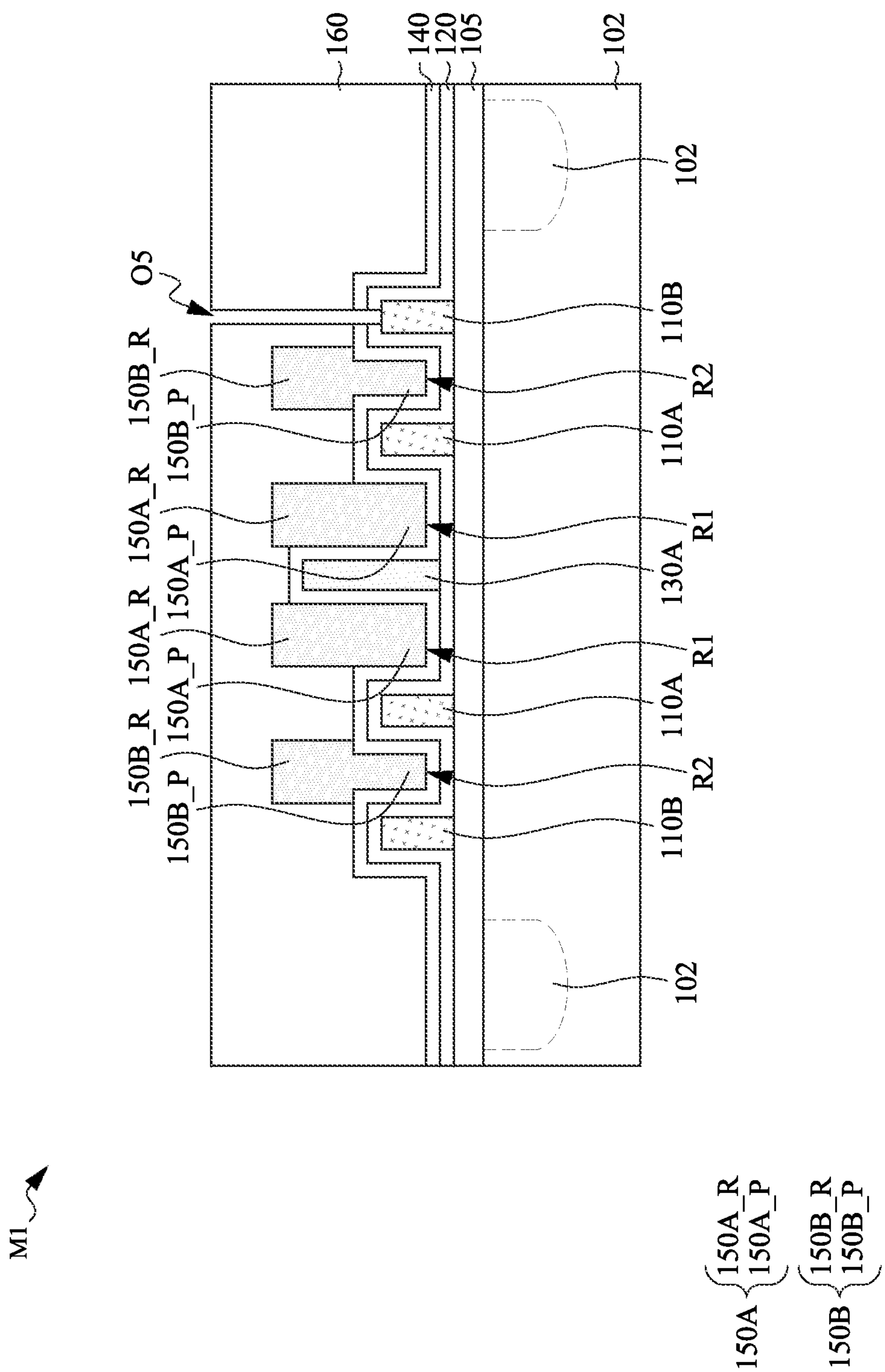

M1
160
140
120
105
102
O4
150B_R
(150B)
O3
O2
150A_R
O1
110B
130B
110A
130A
R1
150A_P
150A
150A_R
150A_P

Fig. 8B



160
140
120
105
102
102
102
O5
150B_R
150B_P
150A_R
150A_P
150A_R
150A_P
150B_R
150B_P
110B
R2
110A
R1
130A
R1
110A
R2
110B
M1
150A
150A_R
150A_P
150B
150B_R
150B_P

Fig. 8C

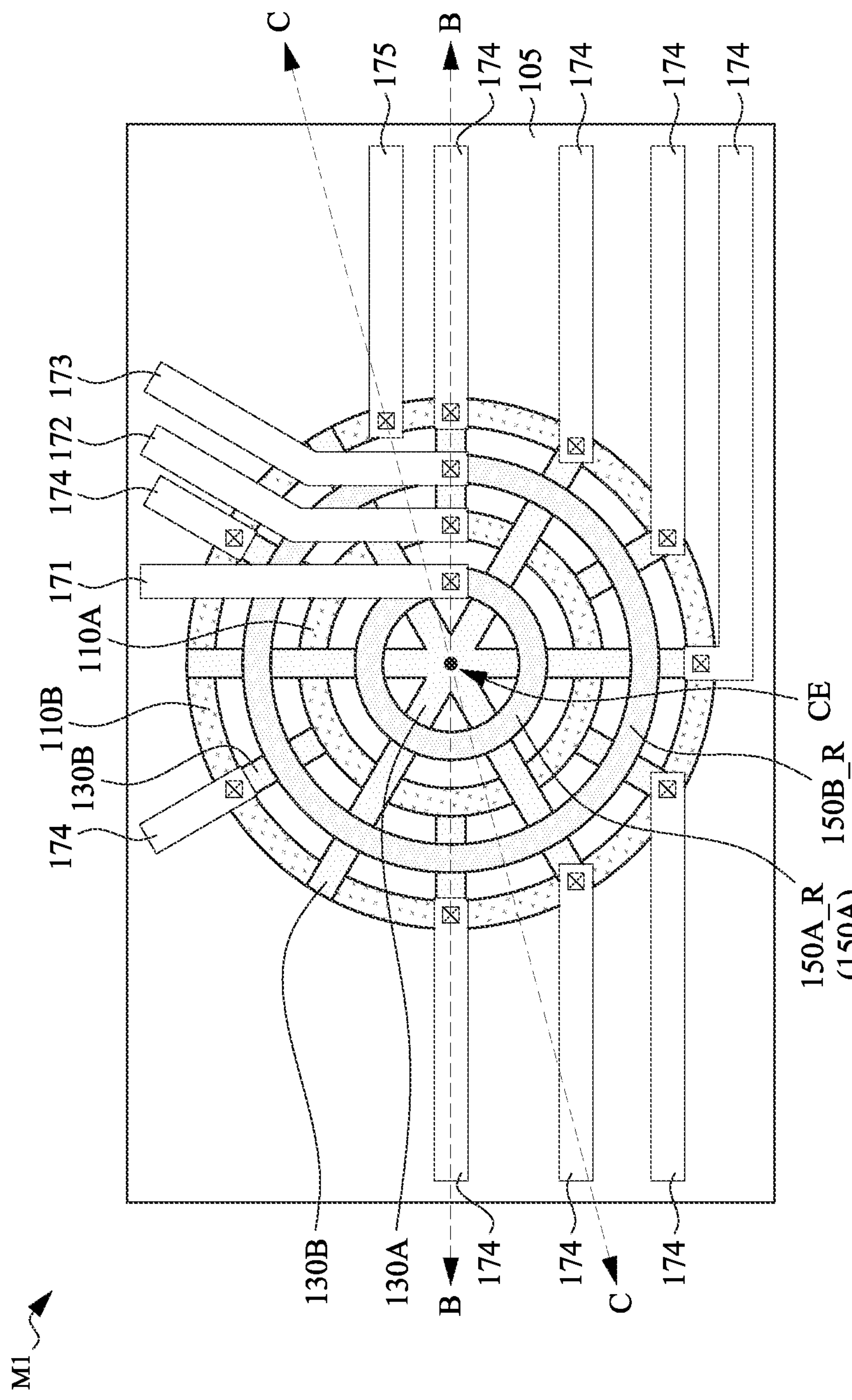

C
B
175
174
105
174
174
174
173
172
174
171
110A
110B
130B
174
CE
150B_R
(150B)
150A_R
(150A)
130B
130A
B
174
174
C
174
M1

Fig. 9A

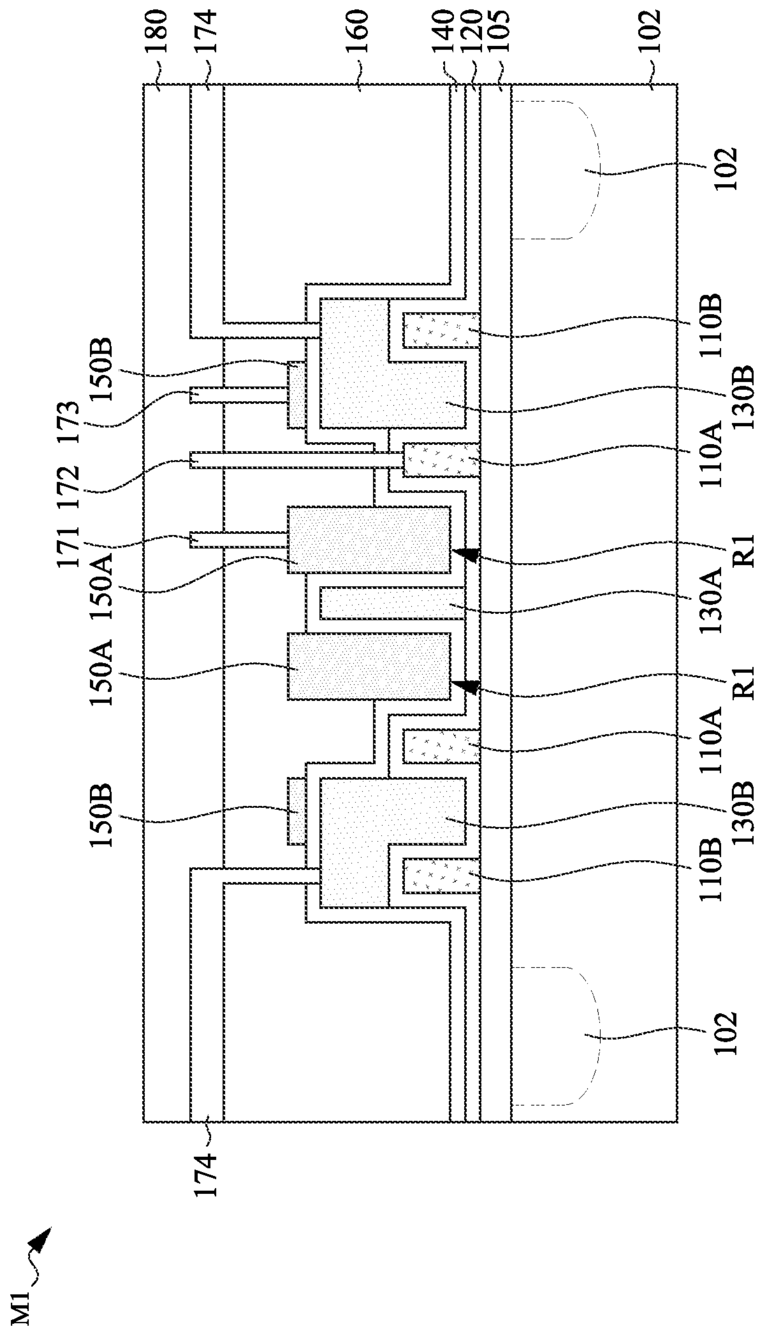

M1
180
174
160
140
120
105
102
102
110B
130B
110A
R1
130A
R1
110A
130B
110B
102
150B
173
172
171
150A
150A
150B
174

Fig. 9B

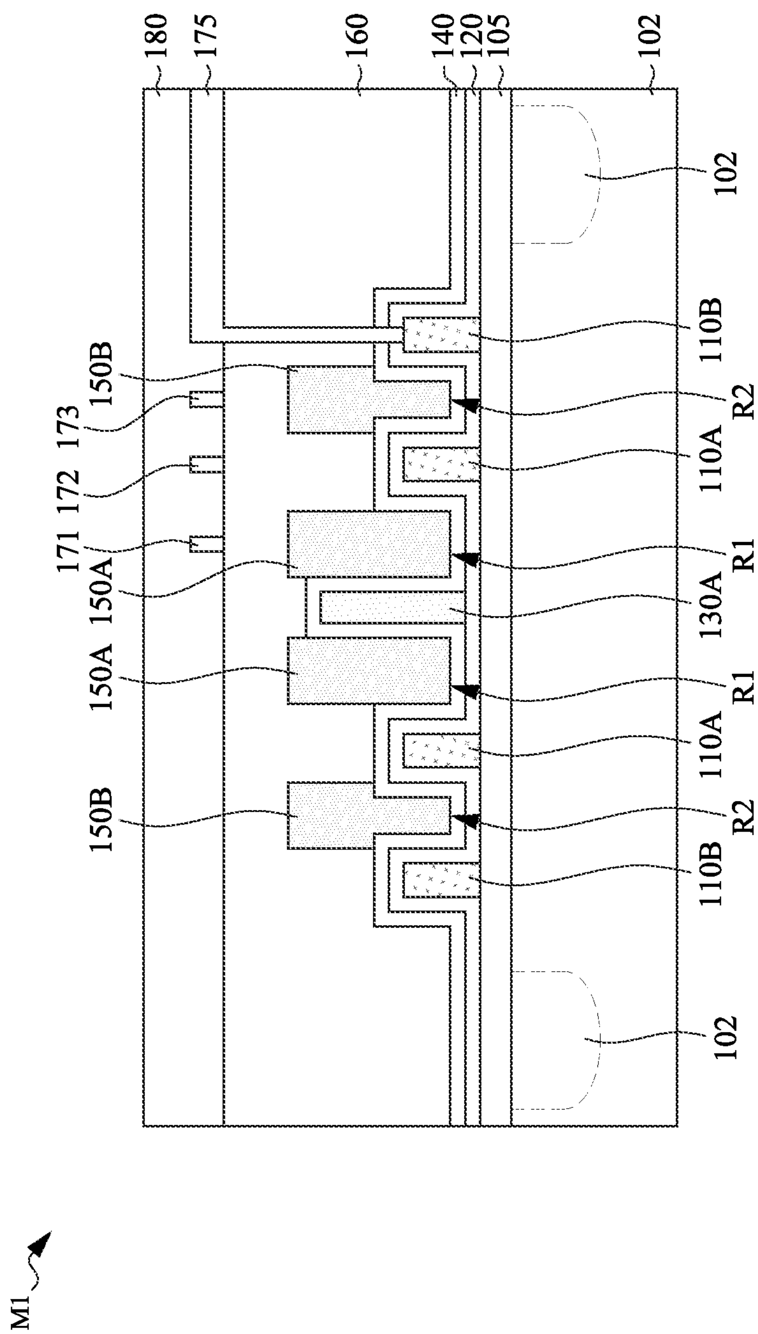

180
175
160
140
120
105
102
102
110B
R2
110A
R1
130A
R1
110A
R2
110B
102
150B
173
172
171
150A
150A
150B
M1

Fig. 9C

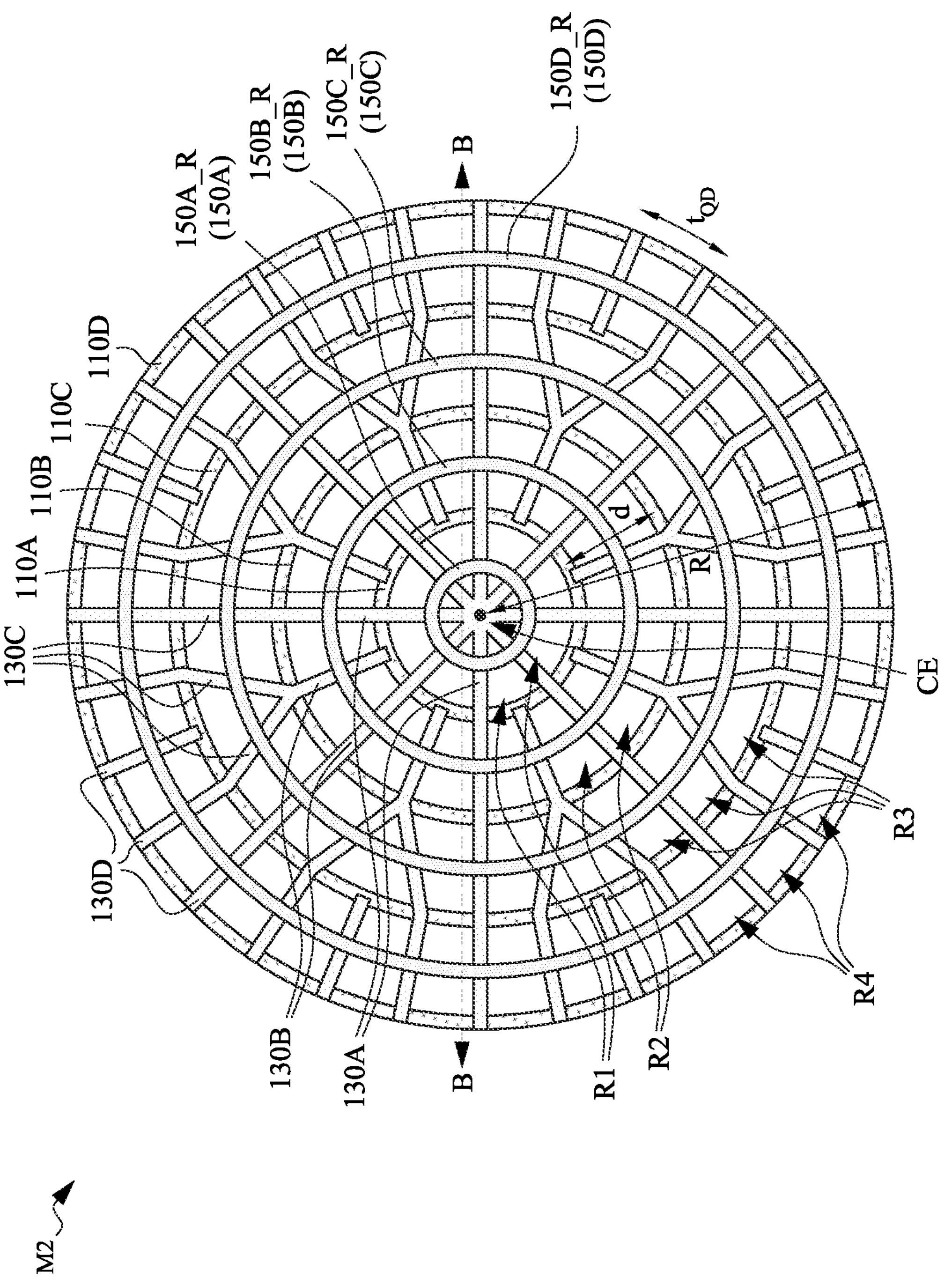

150D_R
(150D)
150C_R
(150C)
150B_R
(150B)
150A_R
(150A)
B
t_{QD}
110D
110C
110B
110A
130C
130D
130B
130A
B
R1
R2
R4
R3
CE
R
d
M2

Fig. 10A

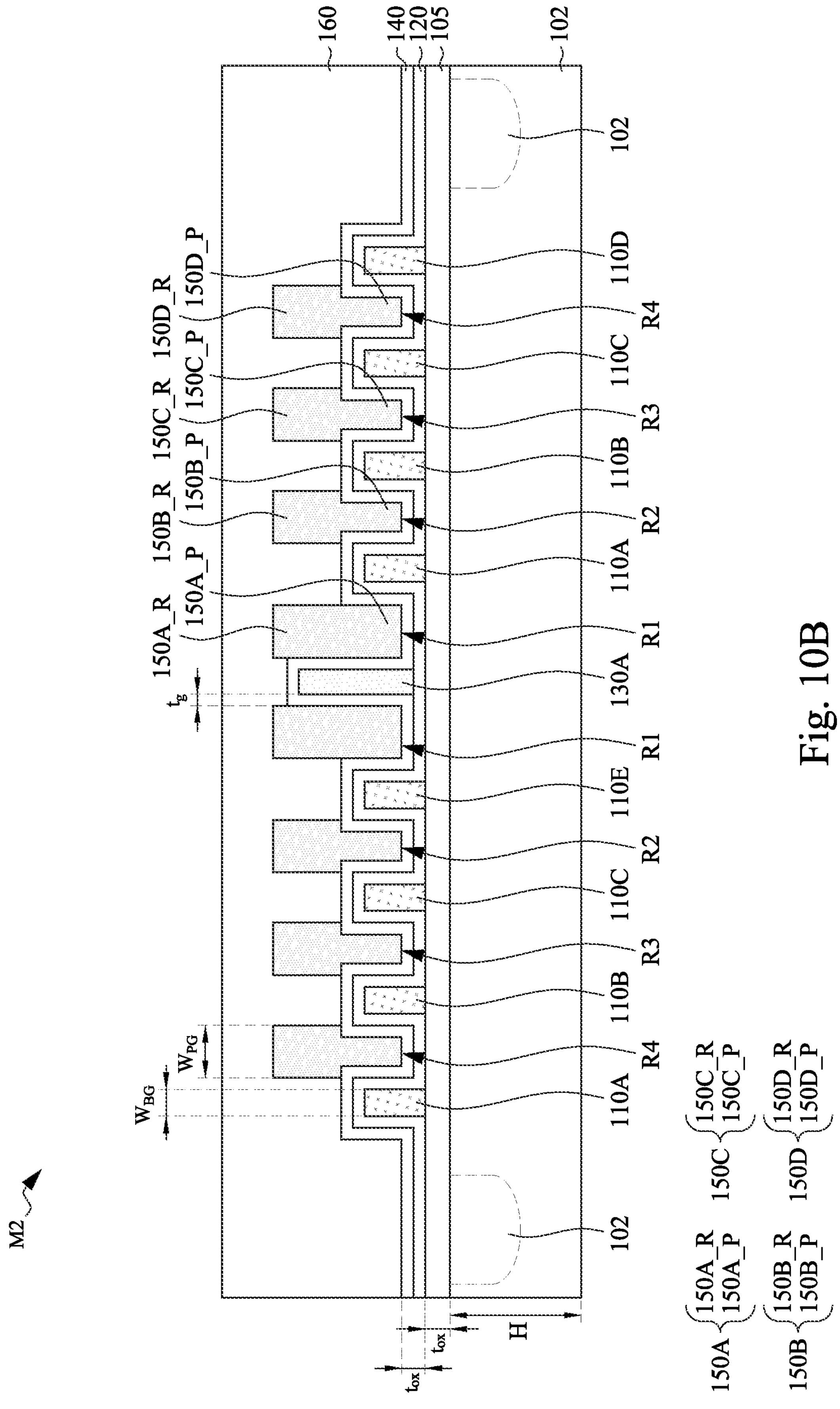

160
140
120
105
102
102
110D
R4
110C
R3
110B
R2
110A
R1
130A
R1
110E
R2
110C
R3
110B
R4
110A
102
150A_R
150A_P
150B_R
150B_P
150C_R
150C_P
150D_R
150D_P
t_g
W_{BG}
W_{PG}
M2
t_{ox}
t_{ox}
H
150A
150B
150C
150D

Fig. 10B

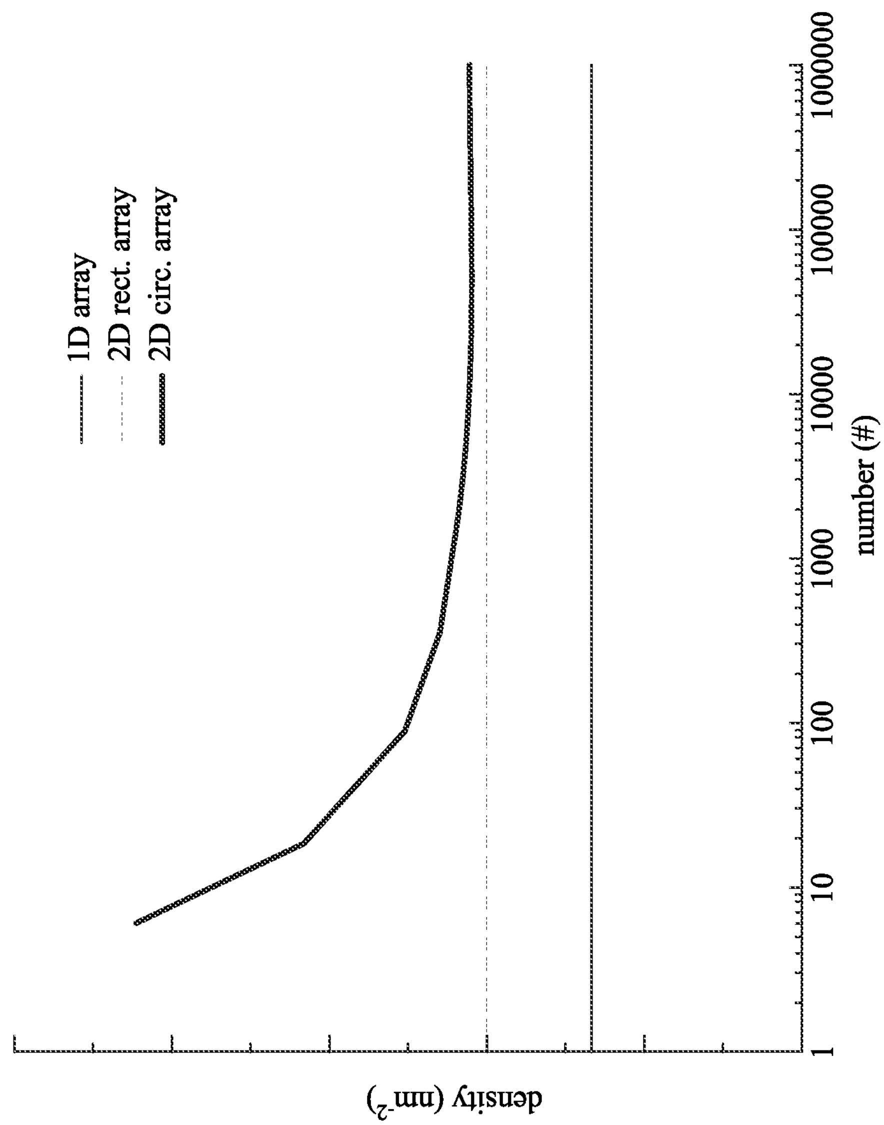

1D array
2D rect. array
2D circ. array
density (nm⁻²)
number (#)
1
10
100
1000
10000
100000
1000000

Fig. 11

MEMORY DEVICE AND METHOD FOR FORMING THE SAME

BACKGROUND

[0001] The semiconductor integrated circuit (IC) industry has experienced rapid growth. Technological advances in IC materials and design have produced generations of ICs. Each generation has smaller and more complex circuits than the previous generation. However, these advances have increased the complexity of processing and manufacturing ICs. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometric size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling-down process generally provides benefits by increasing production efficiency and lowering associated costs. However, since feature sizes continue to decrease, fabrication processes continue to become more difficult to perform. Therefore, it is a challenge to form reliable semiconductor devices at smaller and smaller sizes.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIGS. **1**A to **9**C illustrate a method for manufacturing a memory device at various stages in accordance with some embodiments of the present disclosure.

[0004] FIG. **10**A is a top view of a memory device in accordance with some embodiments of the present disclosure.

[0005] FIG. **10**B is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure.

[0006] FIG. **11** shows simulation results in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0007] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0008] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. As used herein, “around,” “about,” “approximately,” or “substantially” may generally mean within 20 percent, or within 10 percent, or within 5 percent of a given value or range. Numerical quantities given herein are approximate, meaning that the term “around,” “about,” “approximately,” or “substantially” can be inferred if not expressly stated. One skilled in the art will realize, however, that the values or ranges recited throughout the description are merely examples, and may be reduced or varied with the down-scaling of the integrated circuits.

[0009] Quantum computing is capable of solving intractable problems by classical computers, owing to the entanglement and superposition properties of qubits (quantum bits). Si-based spin qubits are promising due to their long decoherence in enriched ^{28}Si quantum dots (QDs) and compatibility with Si very large-scale integration (VLSI) technology. For practical applications, the number of qubits (or QDs) needs to be as large as possible. Thus, the architecture of QDs is critical for large-scale qubits. 1D linear QD arrays are commonly used for large-scale qubits. In the present disclosure, a novel 2D architecture of QD arrays with a larger density is provided, enabling qubit a large-scale spin qubit system compatible with VLSI technologies.

[0010] FIGS. **1**A to **9**C illustrate a method for manufacturing a memory device at various stages in accordance with some embodiments of the present disclosure. In greater detail, FIGS. **1**A, **2**A, **3**A, **4**A, **5**A, **6**A, **7**A, **8**A, and **9**A are top views of a memory device. FIGS. **1**B, **2**B, **3**B, **4**B, **5**B, **6**B, **7**B, **8**B, and **9**B are cross-sectional views along line B-B of FIGS. **1**A, **2**A, **3**A, **4**A, **5**A, **6**A, **7**A, **8**A, and **9**A, respectively. FIGS. **4**C, **5**C, **6**C, **7**C, **8**C, and **9**C are cross-sectional views along line C-C of FIGS. **4**A, **5**A, **6**A, **7**A, **8**A, and **9**A, respectively. It is noted that some elements in the cross-sectional views are omitted in the top views for simplicity. FIG. **6**D is a schematic view of a plunger gate structure of FIG. **6**A. It is noted that some elements in the cross-sectional views are omitted in the top views for simplicity.

[0011] Although the views shown in FIGS. **1**A to **9**C are described with reference to a method, it will be appreciated that the structures shown in FIGS. **1**A to **9**C are not limited to the method but rather may stand alone separate of the method. Although FIGS. **1**A to **9**C are described as a series of acts, it will be appreciated that these acts are not limiting in that the order of the acts can be altered in other embodiments, and the methods disclosed are also applicable to other structures. In other embodiments, some acts that are illustrated and/or described may be omitted in whole or in part.

[0012] Reference is made to FIGS. **1**A and **1**B. Shown there is a memory device **M1**. The memory device **M1** includes a substrate **100**. The substrate **100** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like. The substrate **100** may be a wafer, such as a silicon wafer. Generally, an SOI substrate includes a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **100** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and/or GaInAsP; or combinations thereof.

[0013] Source/drain regions **102** are formed in the substrate **100**. The source/drain regions **102** may be doped regions in the substrate **100**, and may be formed using suitable process, such as an implantation process. The doping species may include p-type dopants, such as boron or BF_2; n-type dopants, such as phosphorus or arsenic; and/or other suitable dopants including combinations thereof. In some embodiments where electrons are used as carriers, the source/drain regions **102** may be n-type doped regions, and the substrate **100** may be p-type substrate. On the other hand, when holes are used as carriers, the source/drain regions **102** may be p-type doped regions, and the substrate **100** may be n-type substrate.

[0014] A gate dielectric layer **105** is formed over the substrate **100**. In some embodiments, the gate dielectric layer **105** may include oxide, such as aluminum oxide (Al_2O_3), silicon oxide (SiO_2), or the like. The gate dielectric layer **105** may also include high-k dielectric material, such as HfO_2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO_2—Al_2O_3) alloy, other suitable high-k dielectric materials, and/or combinations thereof. The gate dielectric layer **105** may be formed using suitable deposition process, such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), or other suitable deposition processes.

[0015] A gate layer **111** is formed over the gate dielectric layer **105**. In some embodiments, the gate layer **111** may be made of conductive material, such as metal, and can also be referred to as a metal layer. In some embodiments, the gate layer **111** may include tungsten (W), aluminum (Al), copper (Cu), gold (Au), chromium (Cr), or the like. In other embodiments, the gate layer **111** may include titanium (Ti), titanium nitride (TiN), or the like. The gate layer **111** may be formed using suitable deposition process, such as sputtering, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or other suitable deposition processes.

[0016] Reference is made to FIGS. **2**A and **2**B. The gate layer **111** is patterned to form a barrier gate **110**A and a barrier gate **110**B separated from each other. As shown in the top view of FIG. **2**A, each of the barrier gates **110**A and **110**B includes an annular top profile (or a circular-ring shape top profile). In some embodiments, the barrier gates **110**A and **110**B are arranged concentrically with respect to a center CE, in which the barrier gate **110**B surrounds the barrier gate **110**A. In some embodiments, the barrier gates **110**A and **110**B may also be referred to as annular barrier gates.

[0017] In some embodiments, the gate layer **111** may be patterned using suitable photolithography process. For example, a patterned mask (e.g., photoresist) layer is formed over the gate layer **111**, the patterned mask may include openings that define the profiles of the barrier gates **110**A and **110**B, an etching process is performed to remove portions of the gate layer **111** through the openings of the patterned mask, and the patterned mask is then removed once the etching process is complete. In some embodiments, the etching process may be reactive ion etching (RIE) process, or may be other suitable etching process.

[0018] Reference is made to FIGS. **3**A and **3**B. A gate dielectric layer **120** is formed over the substrate **100** and lining the underlying structure. In greater detail, the gate dielectric layer **120** is deposited in a conformal manner, such that the gate dielectric layer **120** lines the top surface of the gate dielectric layer **105**, and lines the top surface and opposite sidewalls of each of the barrier gates **110**A and **110**B. In some embodiments, the gate dielectric layer **120** may include oxide, such as aluminum oxide (Al_2O_3), silicon oxide (SiO_2), or the like. The gate dielectric layer **120** may also include high-k dielectric material, such as HfO_2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO_2—Al_2O_3) alloy, other suitable high-k dielectric materials, and/or combinations thereof. The gate dielectric layer **120** may be formed using suitable deposition process, such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), or other suitable deposition processes.

[0019] A gate layer **131** is formed over the gate dielectric layer **120**. In greater detail, the gate layer **131** may cover the barrier gates **110**A and **110**B, and may fill the spaces adjacent to the barrier gates **110**A and **110**B. For example, the gate layer **131** fills the circular region within the barrier gate **110**A, the ring shape region between the barrier gates **110**A and **110**B, and the region outside the barrier gate **110**B. In some embodiments, the gate layer **131** is deposited such that the top surface of the gate layer **131** is higher than the top surfaces of the barrier gates **110**A and **110**B. In some embodiments, the top surfaces of the barrier gates **110**A and **110**B is lower than the top surface of the gate layer **131** and is higher than the bottom surfaces of the gate layer **131**.

[0020] In some embodiments, the gate layer **131** may be made of conductive material, such as metal, and can also be referred to as a metal layer. In some embodiments, the gate layer **131** may include tungsten (W), aluminum (Al), copper (Cu), gold (Au), chromium (Cr), or the like. In other embodiments, the gate layer **131** may include titanium (Ti), titanium nitride (TiN), or the like. The gate layer **131** may be formed using suitable deposition process, such as sputtering, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or other suitable deposition processes.

[0021] Reference is made to FIGS. **4**A, **4**B, and **4**C. The gate layer **131** is patterned to form barrier gates **130**A and **130**B. Here, the barrier gates **130**A may be referred to as the portions of the patterned gate layer **131** extend from the center CE to the barrier gate **110**A. The barrier gates **130**B may be referred to as the portions of the patterned gate layer **131** extend from the barrier gate **110**A to the barrier gate **110**B. In some embodiments, the barrier gates **130**A are connected with each other at the center CE. As shown in the top view of FIG. **4**A, each of the barrier gates **130**A and **130**B includes a linear top profile (or a bar-shape top profile). In some embodiments, each of the barrier gates **130**A and **130**B are arranged radially with respect to the center CE. Accordingly, the barrier gates **130**A and **130**B can also be referred to as radial gates or linear gates.

[0022] In some embodiments, parts of the barrier gates **130**B are connected with the barrier gates **130**A, while parts of the barrier gates **130**B are free of connection with the barrier gates **130**A. As an example of FIG. **4**A, if one barrier gate **130**B is connected with a respective barrier gate **130**A, two barrier gates **130**B on opposite sides of the one barrier gate **130**B are free of connection with the barrier gates **130**A. Similarly, if one barrier gate **130**B is free of connection with the barrier gates **130**A, two barrier gates **130**B on opposite sides of the one barrier gate **130**B are connected with the respective barrier gates **130**A. As a result, the number of the barrier gates **130**B is greater than the number of the barrier gates **130**A.

[0023] After the barrier gates **130**A and **130**B are formed, the barrier gates **110**A and **110**B and the barrier gates **130**A and **130**B collectively define several cavities R**1** and R**2**. The cavities R**1** are arranged annually with respect to the center CE, and the cavities R**2** are arranged annually with respect to the center CE, respectively. In greater detail, the cavities R**1** are arranged along a first ring, and the cavities R**2** are arranged along a second ring that surrounds the first ring. In some embodiments, the first and second rings are arranged concentrically with respect to the center CE.

[0024] As shown in the top view of FIG. **4**A, each of the cavities R**1** is defined by two adjacent barrier gates **130**A and the barrier gate **110**A. As a result, each of the cavities R**1** has a circular sector top profile. That is, the top profile of each cavity R**1** is defined by three sides. On the other hand, each of the cavities R**2** is defined by the barrier gate **110**A, the barrier gate **110**B, and two adjacent barrier gates **130**B. As a result, each of the cavities R**2** has an annular sector top profile. That is, the top profile of each cavity R**2** is defined by four sides.

[0025] Reference is made to FIGS. **5**A, **5**B, and **5**C. A gate dielectric layer **140** is formed over the substrate **100** and lining the underlying structure. In greater detail, the gate dielectric layer **140** is deposited in a conformal manner, such that the gate dielectric layer **140** lines the top surface of the gate dielectric layer **120**, and lines the top surface and opposite sidewalls of each of the barrier gates **130**A and **130**B. In some embodiments, the gate dielectric layer **140** may include oxide, such as aluminum oxide (Al_2O_3), silicon oxide (SiO_2), or the like. The gate dielectric layer **120** may also include high-k dielectric material, such as HfO_2, HfSiO, HfSiON, HfTaO, HfTiO, HfZrO, zirconium oxide, aluminum oxide, titanium oxide, hafnium dioxide-alumina (HfO_2—Al_2O_3) alloy, other suitable high-k dielectric materials, and/or combinations thereof. The gate dielectric layer **140** may be formed using suitable deposition process, such as atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), or other suitable deposition processes.

[0026] A gate layer **151** is formed over the gate dielectric layer **140**. In greater detail, the gate layer **151** may cover the barrier gates **110**A, **110**B, **130**A, and **130**B, and may fill the cavities R**1** and R**2**. In some embodiments, the gate layer **151** is deposited such that the top surface of the gate layer **151** is higher than top surfaces of the barrier gates **130**A and **130**B. In some embodiments, the top surfaces of the barrier gates **110**A and **110**B is lower than the top surface of the gate layer **151** and is higher than the bottom surfaces of the gate layer **151**. Similarly, the top surfaces of the barrier gates **130**A and **130**B is lower than the top surface of the gate layer **151** and is higher than the bottom surfaces of the gate layer **151**.

[0027] In some embodiments, the gate layer **151** may be made of conductive material, such as metal, and can also be referred to as a metal layer. In some embodiments, the gate layer **151** may include tungsten (W), aluminum (Al), copper (Cu), gold (Au), chromium (Cr), or the like. In other embodiments, the gate layer **151** may include titanium (Ti), titanium nitride (TiN), or the like. The gate layer **151** may be formed using suitable deposition process, such as sputtering, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or other suitable deposition processes. In some embodiments, the gate layers **111**, **131**, and **151** may include a same conductive material.

[0028] Reference is made to FIGS. **6**A, **6**B, **6**C, and **6**D. The gate layer **151** is patterned to form plunger gate structures **150**A and **150**B separated from each other. The plunger gate structures **150**A and **150**B are arranged concentrically with respect to the center CE, in which the plunger gate structure **150**B surrounds the plunger gate structure **150**A. In some embodiments, the plunger gate structures **150**A and **150**B may be electrically isolated from the barrier gates **110**A, **110**B, **130**A, and **130**B through dielectric material, such as the gate dielectric layers **120** and **140** as shown in FIGS. **6**B and **6**C.

[0029] FIG. **6**D is a schematic view of the plunger gate structure **150**A, in which each of the plunger gate structure **150**A include a ring structure **150**A_R and a plurality of plunger gates **150**A_P extends downward from the ring structure **150**A_R. In greater detail, the plunger gates **150**A_P may be referred to as the portions of the gate layer **151** that fill the cavities R**1** as described above, and the ring structure **150**A_R may be referred to as the portion of the patterned gate layer **151** above the plunger gates **150**A_P. The ring structure **150**A_R has an annular top profile that crosses each of the barrier gates **130**A, and may be connected with the plunger gates **150**A_P filled in the cavities R**1**. That is, the plunger gates **150**A_P may be electrically connected with each other through the ring structure **150**A_R.

[0030] With respect to the plunger gate structure **150**B, each of the plunger gate structure **150**B include a ring structure **150**B_R and a plurality of plunger gates **150**B_P extends downward from the ring structure **150**B_R. In greater detail, the plunger gates **150**B_P may be referred to as the portions of the gate layer **151** that fill the cavities R**2** as described above, and the ring structure **150**B_R may be referred to as the portion of the patterned gate layer **151** above the plunger gates **150**B_P. The ring structure **150**B_R has an annular top profile that crosses each of the barrier gates **130**B, and may be connected with the plunger gates **150**B_P filled in the cavities R**2**. That is, the plunger gates **150**B_P may be electrically connected with each other through the ring structure **150**B_R.

[0031] Although not shown in FIG. **6**A, it is noted that the plunger gates **150**A_P that fill the cavities R**1** may inherit the profiles of the respective cavities R**1**. Accordingly, the plunger gates **150**A_P may include similar top profiles as the respective cavities R**1** described in FIG. **4**A. Similarly, the plunger gates **150**B_P that fill the cavities R**2** may inherit the profiles of the respective cavities R**2**. Accordingly, the plunger gates **150**B_P may include similar top profiles as the respective cavities R**2** described in FIG. **4**A. It is noted that although the cavities R1 and R2 are lined with the gate dielectric layers **120** and **140**, the gate dielectric layers **120** and **140** are both deposited using conformal deposition methods, and thus the gate dielectric layers **120** and **140** would not affect the profiles of the cavities R1 and R2 defined by the barrier gates **110**A and **110**B and the barrier gates **130**A and **130**B. Detailed description related to the profiles of the cavities R1 and R2 have been discuss above, and will not be repeated for brevity.

[0032] Referring to FIG. **6**A, it can be seen that the barrier gates **110**A and **110**B, the ring structure **150**A_R of the plunger gate structure **150**A, and the ring structure **150**B_R of the plunger gate structure **150**B include annular top profiles (or circular-ring shape top profiles), and are arranged concentrically with respect to the center CE. Moreover, the barrier gate **110**B surrounds the ring structure **150**B_R of the plunger gate structure **150**B, the ring structure **150**B_R of the plunger gate structure **150**B surrounds the barrier gate **110**A, and the barrier gate **110**A surrounds the ring structure **150**A_R of the plunger gate structure **150**A.

[0033] Reference is made to FIGS. **7**A, **7**B, and **7**C. A passivation layer **160** is formed over the substrate **100** and covering the plunger gate structures **150**A and **150**B. In some embodiments, the passivation layer **160** may include silicon oxide, silicon nitride, silicon oxynitride, tetraethoxysilane (TEOS), phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), low-k dielectric material, and/or other suitable dielectric materials. Examples of low-k dielectric materials include, but are not limited to, fluorinated silica glass (FSG), carbon doped silicon oxide, amorphous fluorinated carbon, parylene, bis-benzocyclobutenes (BCB), or polyimide. The passivation layer **160** may be formed using, for example, CVD, ALD, spin-on-glass (SOG) or other suitable techniques. In some embodiments, the passivation layer **160** may include different dielectric material than the gate dielectric layers **105**, **120**, and **140**. For example, the passivation layer **160** may include silicon oxide (SiO_2), and the gate dielectric layers **105**, **120**, and **140** may include aluminum oxide (Al_2O_3).

[0034] Reference is made to FIGS. **8**A, **8**B, and **8**C. The passivation layer **160** is patterned to form via openings O**1**, O**2**, O**3**, O**4**, and O**5** in the passivation layer **160**. In some embodiments, the via opening O**1** extends through the passivation layer **160** and exposes the top surface of the plunger gate structure **150**A. The via opening O**2** extends through the passivation layer **160**, the gate dielectric layer **140**, and the gate dielectric layer **120**, and exposes the top surface of the barrier gate **110**A. The via opening O**3** extends through the passivation layer **160** and exposes the top surface of the plunger gate structure **150**B. The via openings O**4** extend through the passivation layer **160** and the gate dielectric layer **140**, and expose the top surfaces of the respective barrier gates **130**B. The via opening O**5** extends through the passivation layer **160** and the gate dielectric layers **120** and **140**, and exposes the top surface of the barrier gate **110**B (see FIG. **8**C).

[0035] The via openings O**1**, O**2**, O**3**, O**4**, and O**5** may be formed by, for example, forming a patterned mask (e.g., photoresist) layer over the passivation layer **160**, the patterned mask may include openings that define the positions of the via openings O**1**, O**2**, O**3**, O**4**, and O**5**, performing an etching process to remove portions of the passivation layer **160** and the gate dielectric layers **120** and **140** exposed through the openings of the patterned mask, and then removing the patterned mask once the etching process is complete. In some embodiments, the etching process may be reactive ion etching (RIE) process, or may be other suitable etching process.

[0036] Reference is made to FIGS. **9**A, **9**B, and **9**C. Contacts **171**, **172**, **173**, **174**, and **175** are formed over the passivation layer **160** and fill the respective via openings O**1**, O**2**, O**3**, O**4**, and O**5**. In greater detail, the contact **171** has a metal line portion over the passivation layer **160** and a via portion filling the opening O**1**, such that the contact **171** is electrically connected to the plunger gate structure **150**A. The contact **172** has a metal line portion over the passivation layer **160** and a via portion filling the opening O**2**, such that the contact **172** is electrically connected to the barrier gate **110**A. The contact **173** has a metal line portion over the passivation layer **160** and a via portion filling the opening O**3**, such that the contact **173** is electrically connected to the plunger gate structure **150**B. The contacts **174** each has a metal line portion over the passivation layer **160** and a via portion filling the respective opening O**4**, such that the contacts **174** are electrically connected to the respective barrier gates **130**B. The contact **175** has a metal line portion over the passivation layer **160** and a via portion filling the opening O**5**, such that the contact **175** is electrically connected to the barrier gate **110**B. In some embodiments, because some of the barrier gates **130**A are connected with the barrier gates **130**B, the barrier gates **130**A may also be electrically connected to the contacts **174** through the respective barrier gates **130**B.

[0037] The contacts **171**, **172**, **173**, **174**, and **175** may be formed by, for example, forming a patterned mask (e.g., photoresist) layer over the passivation layer **160**, the patterned mask may include openings that define the positions and profiles of the contacts **171**, **172**, **173**, **174**, and **175**, depositing a conductive material over the substrate **100** and filling the openings of the patterned mask and the via openings O**1**, O**2**, O**3**, O**4**, and O**5** in the passivation layer **160**, and then performing a lift-off process to remove the patterned mask, leaving the portions of the conductive material within the openings of the patterned mask and the via openings O**1**, O**2**, O**3**, O**4**, and O**5** remaining over the substrate **100** as the contacts **171**, **172**, **173**, **174**, and **175**.

[0038] In some embodiments, the contacts **171**, **172**, **173**, **174**, and **175** may include gold (Au), chromium (Cr), copper (Cu), or the like. The contacts **171**, **172**, **173**, **174**, and **175** may be formed using suitable deposition process, such as sputtering, chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or other suitable deposition processes.

[0039] After the contacts **171** to **175** are formed, a passivation layer **180** is formed over the passivation layer **160** and covering the contacts **171** to **175**. The passivation layer **180** may include similar material as the passivation layer **160**, and thus relevant details will not be repeated for brevity.

[0040] FIG. **10**A is a top view of a memory device in accordance with some embodiments of the present disclosure. FIG. **10**B is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure. In greater detail, FIG. **10**B is a cross-sectional view along line B-B of FIG. **10**A. It is noted that some elements of FIGS. **10**A and **10**B are similar with those described with respect to FIGS. **1**A to **9**C, such elements are labeled the same and relevant details will not be repeated for brevity.

[0041] The Embodiment of FIGS. 10A and 10B is different from the embodiment described with respect to FIGS. 1A to 9C, in that the memory device of FIGS. 10A and 10B further includes barrier gates 110C and 110D, barrier gates 130C and 130D, and plunger gate structures 150C and 150D.

[0042] With respect to the barrier gates 110C and 110D, the barrier gates 110C and 110D each includes an annular top profile (or circular-ring shape), which is similar to the barrier gates 110A and 110B as described above. In some embodiments, the barrier gates 110A, 110B, 110C, and 110D are arranged concentrically with respect to the center CE, in which the barrier gate 110D surrounds the barrier gate 110C, the barrier gate 110C surrounds the barrier gate 110B, and the barrier gate 110B surrounds the barrier gate 110A. In some embodiments, the barrier gates 110A to 110D may also be referred to as annular barrier gates.

[0043] The barrier gates 110C and 110D may be formed together with the barrier gates 110A and 110B during the process of patterning the gate layer 111 as described in FIGS. 2A and 2B. After the barrier gates 110A to 110D are formed, the gate dielectric layer 120 is formed over the substrate 100 and lining the exposed surfaces of the barrier gates 110A to 110D as similar to those described in FIGS. 3A and 3B.

[0044] With respect to the barrier gates 130C and 130D, the barrier gates 130C extend from the barrier gate 110B to the barrier gate 110C, and the barrier gates 130D extend from the barrier gate 110C to the barrier gate 110D. As shown in the top view of FIG. 10A, each of the barrier gates 130C and 130D includes a linear top profile (or bar-shape). In some embodiments, each of the barrier gates 130D and some of the barrier gates 130C are arranged radially with respect to the center CE.

[0045] The barrier gates 130C and 130D may be formed together with the barrier gates 130A and 130B during the process of patterning the gate layer 131 as described in FIGS. 4A, 4B, and 4C. After the barrier gates 130A to 130D are formed, the gate dielectric layer 140 is formed over the substrate 100 and lining the exposed surfaces of the barrier gates 130A to 130D as similar to those described in FIGS. 5A, 5B, and 5C.

[0046] After the barrier gates 130A to 130D are formed, several cavities R1, R2, R3 and R4 are formed. In greater detail, the barrier gates 110B/110C/110D and the barrier gates 130C/130D collectively define the cavities R3 and R4. The cavities R3 are arranged annually with respect to the center CE, and the cavities R4 are arranged annually with respect to the center CE, respectively. In greater detail, the cavities R3 are arranged along a third ring, and the cavities R4 are arranged along a fourth ring that surrounds the third ring.

[0047] As shown in the top view of FIG. 10A, the barrier gates 130C are connected with the respective barrier gates 130B. However, one barrier gate 130B may be connected with a single barrier gate 130C, while another one barrier gate 130B may also be connected with two barrier gates 130C. This will result in that the cavities R3 have different top profiles. For example, first portions of the cavities R3 are defined by two adjacent barrier gates 130C and the barrier gate 110C. The top profile of the first portions of the cavities R3 is defined by three sides, and has a circular sector top profile. On the other hand, second portions of the cavities R3 are defined by two adjacent barrier gates 130C, the barrier gate 110B, and the barrier gate 110C. The top profile of the second portions of the cavities R3 is defined by four sides, and has an annular sector top profile.

[0048] Each of the cavities R4 is defined by the barrier gate 110C, the barrier gate 110D, and two adjacent barrier gates 130D. As a result, each of the cavities R2 has an annular sector top profile. That is, the top profile of each cavity R4 is defined by four sides. In some embodiments, parts of the barrier gates 130D are connected with the barrier gates 130C, while parts of the barrier gates 130D are free of connection with the barrier gates 130C.

[0049] The plunger gate structures 150A, 150B, 150C, and 150D are arranged concentrically with respect to the center CE, in which the plunger gate structure 150D surrounds the plunger gate structure 150C, the plunger gate structure 150C surrounds the plunger gate structure 150B, and the plunger gate structure 150B surrounds the plunger gate structure 150A.

[0050] The plunger gate structures 150C and 150D may be formed together with the plunger gate structures 150A and 150B during the process of patterning the gate layer 151 as described in FIGS. 6A, 6B, and 6C. After the plunger gate structures 150A to 150D are formed, the passivation layer 160 is formed over the substrate 100 and lining the exposed surfaces of the plunger gate structures 150A to 150D as similar to those described in FIGS. 7A, 7B, and 7C.

[0051] With respect to the plunger gate structure 150C, each of the plunger gate structure 150C include a ring structure 150C_R and a plurality of plunger gates 150C_P extends downward from the ring structure 150C_R. In greater detail, the plunger gates 150C_P may be referred to as the portions of the gate layer 151 that fill the cavities R3, and the ring structure 150C_R may be referred to as the portion of the patterned gate layer 151 above the plunger gates 150C_P. The ring structure 150C_R has an annular top profile that crosses each of the barrier gates 130C, and may be connected with the plunger gates 150C_P filled in the cavities R3. That is, the plunger gates 150C_P may be electrically connected with each other through the ring structure 150C_R.

[0052] With respect to the plunger gate structure 150D, each of the plunger gate structure 150D include a ring structure 150D_R and a plurality of plunger gates 150D_P extends downward from the ring structure 150D_R. In greater detail, the plunger gates 150D_P may be referred to as the portions of the gate layer 151 that fill the cavities R4, and the ring structure 150D_R may be referred to as the portion of the patterned gate layer 151 above the plunger gates 150D_P. The ring structure 150D_R has an annular top profile that crosses each of the barrier gates 130D, and may be connected with the plunger gates 150D_P filled in the cavities R4. That is, the plunger gates 150D_P may be electrically connected with each other through the ring structure 150D_R.

[0053] Although not shown in FIG. 10A, it is noted that the plunger gates 150C_P that fill the cavities R3 may inherit the profiles of the respective cavities R3. Accordingly, the plunger gates 150C_P may include similar top profiles as the respective cavities R3. Similarly, the plunger gates 150D_P that fill the cavities R4 may inherit the profiles of the respective cavities R4. Accordingly, the plunger gates 150D_P may include similar top profiles as the respective cavities R4. It is noted that although the cavities R1 to R4 are lined with the gate dielectric layers 120 and 140, the gate dielectric layers **120** and **140** are both deposited using conformal deposition methods, and thus the gate dielectric layers **120** and **140** would not affect the profiles of the cavities R**1** to R**4** defined by the barrier gates **110**A to **110**D and the barrier gates **130**A to **130**D.

[0054] In operation of the memory device M**2**, the plunger gates **150**A_P to **150**D_P may define qubits (or quantum dot; QD) in the substrate **100**. Stated another way, the locations of the qubits are defined by the barrier gates **110**A to **110**D and the barrier gates **130**A to **130**D. The locations of the qubits correspond to the respective plunger gates **150**A_P to **150**D_P, and thus the plunger gates **150**A_P to **150**D_P may also be referred to as the qubits of the memory device M**2**. The qubits of the memory device M**2** are arranged in a circular array on a center region of the substrate **100**, and the source/drain regions **102** are within a peripheral region of the substrate **100** that surrounds the center region of the substrate **100**. In some embodiments, the plunger gates **150**A_P to **150**D_P may also be referred to as the memory cells of the memory device M**2**.

[0055] With respect to the cross-sectional view of FIG. **10**B. In some embodiments, the thickness H of the substrate **100** is in a range from about 500 μm to about 2000 μm.

[0056] In some embodiments, the thickness tox of the gate dielectric layers **105**, **120**, and **140** is in a range from about 5 nm to about 20 nm. In some embodiments, the height HG of the barrier gates **130**A to **130**D, and the plunger gates **150**A_P to **150**D_P is in a range from about 25 nm to about 50 nm. In some embodiments, the width W_{BG} of the barrier gates **110**A to **110**D and the width W_{PG} of the plunger gates **150**A_P to **150**D_P are in a range from about 20 nm to about 50 nm. In some embodiments, the thickness t_g of the dielectric layer (e.g., dielectric layer **140**) between the overlapping gates (e.g., the barrier gate **130**A and the plunger gates **150**A) is in a range from about 5 nm to about 20 nm. In some embodiments, the critical gate spacing t_{crit} between two plunger gates (e.g., plunger gates **150**A_P and **150**B_P) may be greater than $W_{PG}+2t_{ox}$. In some embodiments, the critical gate spacing t_{crit} between two barrier gates (e.g., barrier gates **110**A and **110**B) may be greater than $W_{BG}+2t_{ox}$.

[0057] With respect to the top view of FIG. **10**A, the number of the plunger gates **150**A_P at the 1^{st} ring may be λ (e.g., 8 in this case). The number of the plunger gates **150**B_P at the 2^{nd} ring may be λ*2 (e.g., 16 in this case). The number of the plunger gates **150**C_P at the 3^{rd} ring may be λ*3 (e.g., 24 in this case). The number of the plunger gates **150**D_P at the 4^{th} ring may be λ*4 (e.g., 32 in this case). That is, the number of the plunger gates at n^{th} ring would be λ*n, in which λ is the number of plunger gates at the 1^{st} ring, and n is positive integer.

[0058] In FIG. **10**A, it can be seen that each plunger gates may include a thickness t_{QD} along circumferential direction. For the 1^{st} ring, the thickness top of each plunger gate (e.g., plunger gate **150**A_P) is 2πd/λ, in which d is the spacing between adjacent two annular barrier gates (e.g., barrier gates **110**A to **110**D). For the n^{th} ring, the thickness t_{QD} of each plunger gate is 2πR/nλ, in which R=n×d (radius of the n^{th} ring). That is, the thickness t_{QD} of each plunger gate is

$$\frac{2\pi(n\times d)}{n\lambda}=\frac{2\pi d}{\lambda}.$$

In some embodiments, the thickness t_{QD} may be large enough, such as greater than the critical gate spacing t_{crit} (e.g., $t_{QD}\geq t_{crit}$). That is,

$$\frac{2\pi d}{\lambda}\geq t_{crit},$$

and thus

$$d\geq\frac{t_{crit}\lambda}{2\pi}.$$

In some embodiments, if λ≥6, the value

$$d=\frac{t_{crit}\lambda}{2\pi}$$

can be chosen to ensure that d is not too small to be fabricated. However, if λ<6, the value of d must be selected as t_{crit}. As a result, the density in 2D circular QD array can be optimized by choosing λ is equal to or greater than 6. In some embodiments, λ is in a range from 6 to 12.

[0059] The density of the qubits (e.g., plunger gates) can be expressed as:

$$\begin{aligned}\frac{(T_1+T_2+\cdots+T_n)}{\pi(R)^2}&=\frac{(\lambda+2\lambda+\cdots+n\lambda)}{\pi(R)^2}\\&=\frac{\lambda(1+2+\cdots+n)}{\pi(R)^2}\\&=\frac{\frac{\lambda n(1+n)}{2}}{\pi(n\times d)^2}\end{aligned}$$

[0060] In some embodiments, when n is a large number, the density of the qubits (e.g., plunger gates) can be expressed as:

$$\begin{aligned}&=\frac{\frac{\lambda n(1+n)}{2}}{\pi(n\times d)^2}\\&\approx\frac{\frac{\lambda n(n)}{2}}{\pi(n\times d)^2}\\&=\frac{\lambda n^2}{2\pi(n\times d)^2}\end{aligned}$$

[0061] FIG. **11** shows simulation results in accordance with some embodiments of the present disclosure. In greater detail, simulation results of three different arrangements of qubits are shown. It can be seen that when the qubits are arranged in a 2-D circular array as provided by the present disclosure, the density of qubits may be higher than qubits arranged in a 1-D linear array or in a 2-D rectangular array. Accordingly, circular gate design enables higher quantum dot density and a more compact device configuration. The design of 2-D qubit array allows for programmable entanglement between quantum dots.

[0062] According to the aforementioned embodiments, it can be seen that the present disclosure offers advantages in fabricating integrated circuits. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. Embodiments of the present disclosure provide a qubit memory device, in which the qubits are arranged in a 2-D circular array. The circular gate design enables higher quantum dot density and a more compact device configuration.

[0063] In some embodiments of the present disclosure, a memory device includes a substrate. First plunger gates are over the substrate and arranged annularly. Second plunger gates are arranged annularly around the first plunger gates. A first annular barrier gate is over the substrate and disposed between the first plunger gates and the second plunger gates.

[0064] In some embodiments, the memory device further includes first linear barrier gates over the substrate, wherein each of the first linear barrier gates separates correspond two of the first plunger gates.

[0065] In some embodiments, the first linear barrier gates extend radially from a top view.

[0066] In some embodiments, the memory device further includes a second annular barrier gate over the substrate and surrounding the first annular barrier gate. Second linear barrier gates extend from the first annular barrier gate to the second annular barrier gate, wherein each of the second linear barrier gates separates correspond two of the second plunger gates.

[0067] In some embodiments, one of the second linear barrier gates is connected with one of the first linear gates.

[0068] In some embodiments, the memory device further includes a first ring structure above the first plunger gates and electrically connecting the first plunger gates with each other.

[0069] In some embodiments, a number of the second plunger gates is greater than a number of the first plunger gates.

[0070] In some embodiments, the number of the second plunger gates is multiple times the number of the first plunger gates.

[0071] In some embodiments of the present disclosure, a memory device includes a substrate. A gate dielectric layer is over the substrate. A first plunger gate structure and a second plunger gate structure are over the gate dielectric layer, wherein in a top view the first and second plunger gate structures are arranged concentrically, and wherein each of the first and second plunger gate structures comprises a ring structure and plunger gates extend downwardly from the ring structure. Barrier gates are over the gate dielectric layer, wherein in a cross-sectional view each of the plunger gates of the first and second plunger gate structures is laterally between adjacent two of the barrier gates.

[0072] In some embodiments, the plunger gates of the first plunger gate structure are arranged annularly, and the plunger gates of the second plunger gate structure are arranged annularly.

[0073] In some embodiments, the barrier gates comprise an annular barrier gate, and the annular barrier gate separates the first plunger gate structure from the second plunger gate structure.

[0074] In some embodiments, the barrier gates comprise linear barrier gates, and each of the linear barrier gates is between two of the plunger gates.

[0075] In some embodiments, top surfaces of the linear barrier gates are higher than a top surface of the annular barrier gate in the cross-sectional view.

[0076] In some embodiments, top surfaces of the first and second plunger gate structures are higher than top surfaces of the barrier gates in the cross-sectional view.

[0077] In some embodiments, the first and second plunger gate structures and the barrier gates are made of conductive materials.

[0078] In some embodiments of the present disclosure, a method includes forming source/drain regions in a substrate; forming a first gate dielectric layer over the substrate; forming barrier gates over the substrate, the barrier gates defining a first set of cavities and a second set of cavities arranged around the first set of cavities, wherein cavities in the first set of cavities are arranged annularly, and cavities in the second set of cavities are arranged annularly; and forming plunger gates in the cavities.

[0079] In some embodiments, forming the barrier gates comprises forming annular barrier gates of the barrier gates over the substrate; and forming linear barrier gates of the barrier gates over the substrate after forming the annular barrier gates of the barrier gates.

[0080] In some embodiments, the method further includes forming a second gate dielectric layer lining the annular barrier gates of the barrier gates prior to forming the linear barrier gates of the barrier gates; and forming a third gate dielectric layer lining the linear barrier gates of the barrier gates prior to forming the plunger gates.

[0081] In some embodiments, forming the plunger gates further comprises forming ring structures above and connected with the plunger gates.

[0082] In some embodiments, the barrier gates and the plunger gates are made of conductive materials.

[0083] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A memory device, comprising:

a substrate;

first plunger gates over the substrate and arranged annularly;

second plunger gates arranged annularly around the first plunger gates; and

a first annular barrier gate over the substrate and disposed between the first plunger gates and the second plunger gates.

2. The memory device of claim **1**, further comprising:

first linear barrier gates over the substrate, wherein each of the first linear barrier gates separates correspond two of the first plunger gates.

3. The memory device of claim **2**, wherein the first linear barrier gates extend radially from a top view.

4. The memory device of claim **2**, further comprising:
a second annular barrier gate over the substrate and surrounding the first annular barrier gate; and
second linear barrier gates extending from the first annular barrier gate to the second annular barrier gate, wherein each of the second linear barrier gates separates correspond two of the second plunger gates.

5. The memory device of claim **4**, wherein one of the second linear barrier gates is connected with one of the first linear barrier gates.

6. The memory device of claim **1**, further comprising a first ring structure above the first plunger gates and electrically connecting the first plunger gates with each other.

7. The memory device of claim **1**, wherein a number of the second plunger gates is greater than a number of the first plunger gates.

8. The memory device of claim **7**, wherein the number of the second plunger gates is multiple times the number of the first plunger gates.

9. A memory device, comprising:
a substrate;
a gate dielectric layer over the substrate;
a first plunger gate structure and a second plunger gate structure over the gate dielectric layer, wherein in a top view the first and second plunger gate structures are arranged concentrically, and wherein each of the first and second plunger gate structures comprises a ring structure and plunger gates extend downwardly from the ring structure; and
barrier gates over the gate dielectric layer, wherein in a cross-sectional view each of the plunger gates of the first and second plunger gate structures is laterally between adjacent two of the barrier gates.

10. The memory device of claim **9**, wherein the plunger gates of the first plunger gate structure are arranged annularly, and the plunger gates of the second plunger gate structure are arranged annularly.

11. The memory device of claim **9**, wherein the barrier gates comprise an annular barrier gate, and the annular barrier gate separates the first plunger gate structure from the second plunger gate structure.

12. The memory device of claim **11**, wherein the barrier gates comprise linear barrier gates, and each of the linear barrier gates is between two of the plunger gates.

13. The memory device of claim **12**, wherein top surfaces of the linear barrier gates are higher than a top surface of the annular barrier gate in the cross-sectional view.

14. The memory device of claim **9**, wherein top surfaces of the first and second plunger gate structures are higher than top surfaces of the barrier gates in the cross-sectional view.

15. The memory device of claim **9**, wherein the first and second plunger gate structures and the barrier gates are made of conductive materials.

16. A method, comprising:
forming source/drain regions in a substrate;
forming a first gate dielectric layer over the substrate;
forming barrier gates over the substrate, the barrier gates defining a first set of cavities and a second set of cavities arranged around the first set of cavities, wherein cavities in the first set of cavities are arranged annularly, and cavities in the second set of cavities are arranged annularly; and
forming plunger gates in the cavities.

17. The method of claim **16**, wherein forming the barrier gates comprises:
forming annular barrier gates of the barrier gates over the substrate; and
forming linear barrier gates of the barrier gates over the substrate after forming the annular barrier gates of the barrier gates.

18. The method of claim **17**, further comprising:
forming a second gate dielectric layer lining the annular barrier gates of the barrier gates prior to forming the linear barrier gates of the barrier gates; and
forming a third gate dielectric layer lining the linear barrier gates of the barrier gates prior to forming the plunger gates.

19. The method of claim **16**, wherein forming the plunger gates further comprises forming ring structures above and connected with the plunger gates.

20. The method of claim **16**, wherein the barrier gates and the plunger gates are made of conductive materials.

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